

DRILLING GEOMECHANICS

From rock, stress & pressure to safer drilling decisions

Awareness-level training for a multidisciplinary drilling audience

Eril S. Lanin, Ph.D.

PT FERG Geosains Indonesia

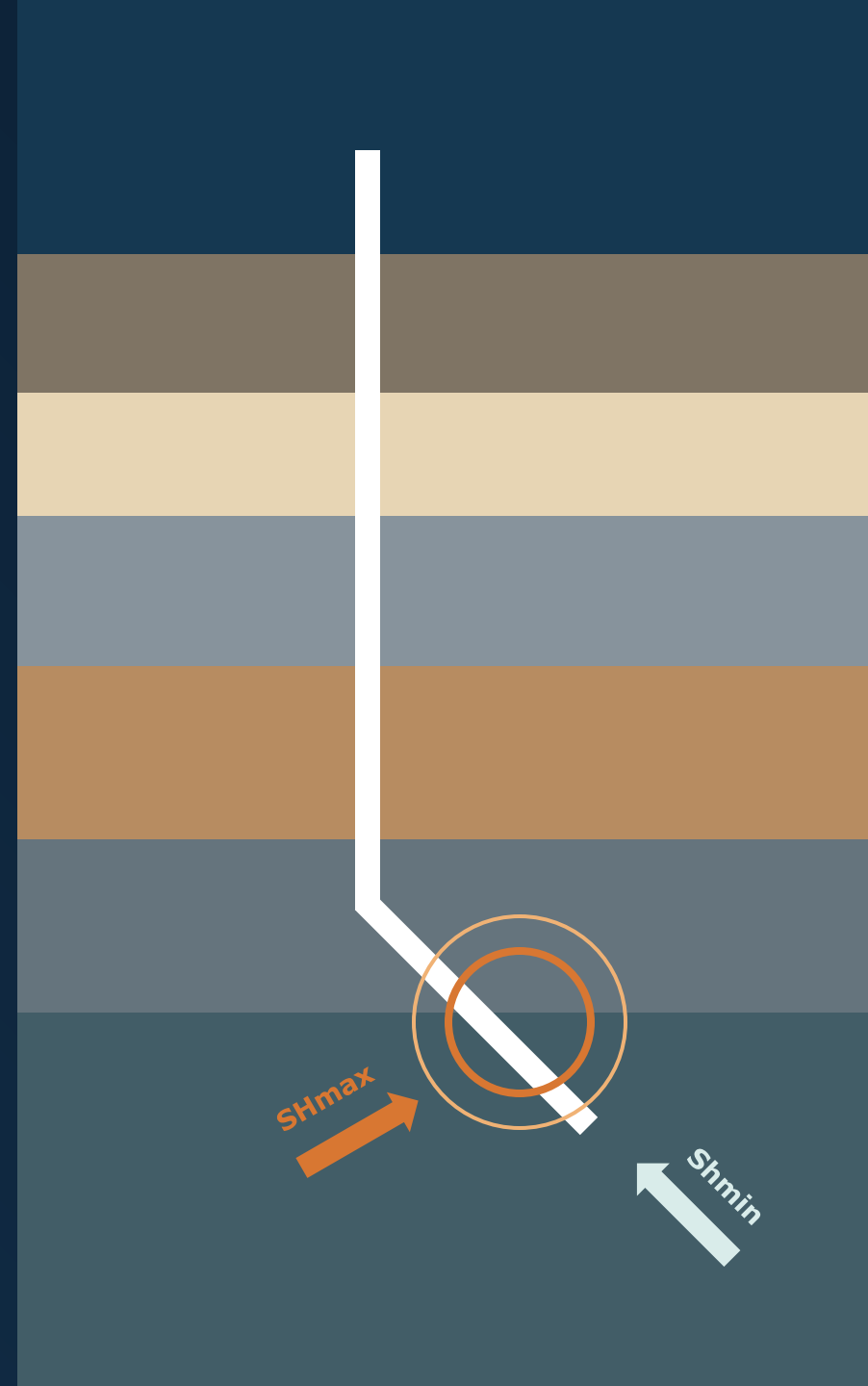


Jakarta Drilling Society • 26 September 2026 • 09:00–15:00



Supported by Hitachi High-Tech

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The organizer's brief

One objective, seven requested topics and an awareness-level audience

OBJECTIVE

Provide participants with an understanding of rock mechanics, in-situ stress, wellbore stability and drilling practices that help prevent wellbore failure and optimize drilling performance.

REQUESTED TOPICS

01 Introduction to Drilling Geomechanics

02 In-Situ Stress & Rock Properties

03 Wellbore Stability Principles

04 Formation Pressure & Fracture Gradient

05 Mud Weight Window & Drilling Fluids

06 Wellbore Failure & Stuck Pipe Prevention

07 Case Studies & Application

AWARENESS LEVEL

Mixed audience: drilling engineers, subsurface professionals, and non-engineering participants

Three questions will guide the day

The organizer's seven topics are retained—but taught through drilling decisions

01

Why does the hole fail?

Why can a deliberately drilled hole enlarge, close, fracture or deteriorate with time?

02

How much pressure is enough?

How do pore pressure, collapse risk and fracture/loss constraints define the operating window?

03

What is the well telling us?

Which observations distinguish a geomechanical problem from hydraulic, chemical or operational causes?

Outcome: recognize the mechanism, identify the missing evidence, and choose the next safe action

One borehole can produce different symptoms

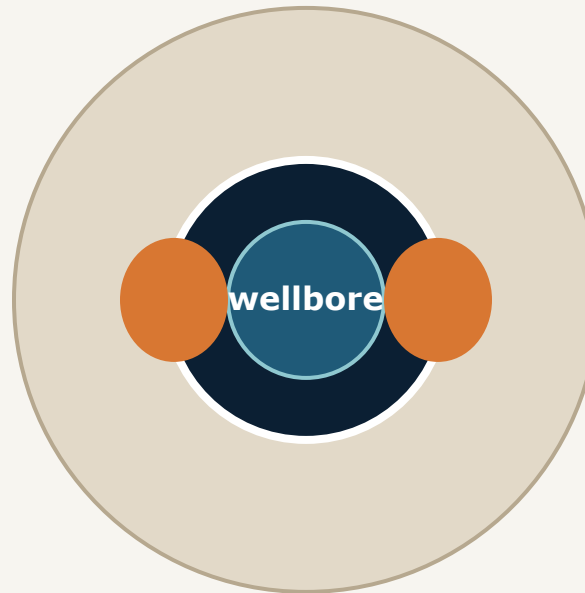
A symptom is evidence—not yet a diagnosis

Hole enlargement

Breakout, washout or time-dependent deterioration can enlarge the hole.

Tight hole / pack-off

Cavings, fill and poor transport can reduce annular clearance.



Fracture / losses

Induced fractures or weak natural features can create fluid-loss pathways.

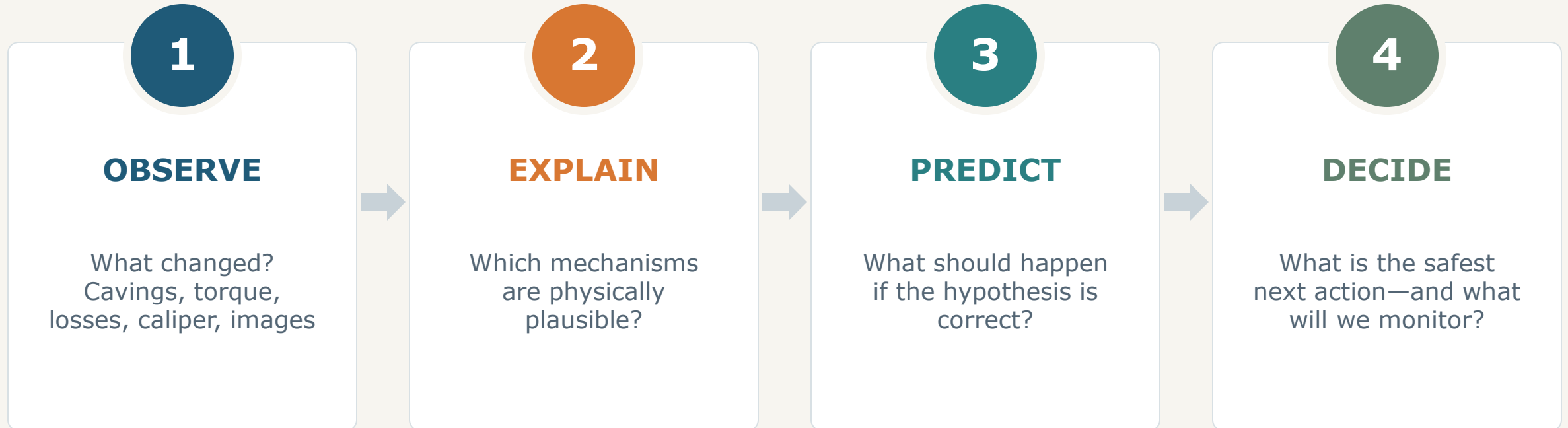
Stuck pipe / NPT

Several mechanisms can converge on the same costly operational outcome.

**Do not prescribe the remedy
before identifying the mechanism**

The drilling-geomechanics decision loop

The same reasoning loop will be used throughout the course



Good geomechanics is a calibrated reasoning process—not a one-time model output

PART I

First, understand the system

The pre-drill model combines far-field conditions, rock behavior and well design

Focus for this section

In-situ stress, pore pressure, rock properties, geology, well geometry and drilling conditions



Borehole response reflects the complete system

Drilling behavior emerges from an interacting system

No single input explains the borehole response by itself



Stress, pore pressure and strength answer different questions

Confusing them leads directly to incorrect drilling interpretations

STRESS

What is loading the rock?

MPa or psi

- S_v , S_{Hmax} , S_{Hmin}
- Total and effective stress
- Magnitude + direction

PORE PRESSURE

What pressure acts in connected pores?

MPa, psi, or gradient

- Hydrostatic / normal
- Overpressure / depletion
- Controls effective stress

ROCK STRENGTH

How much loading can the rock resist?

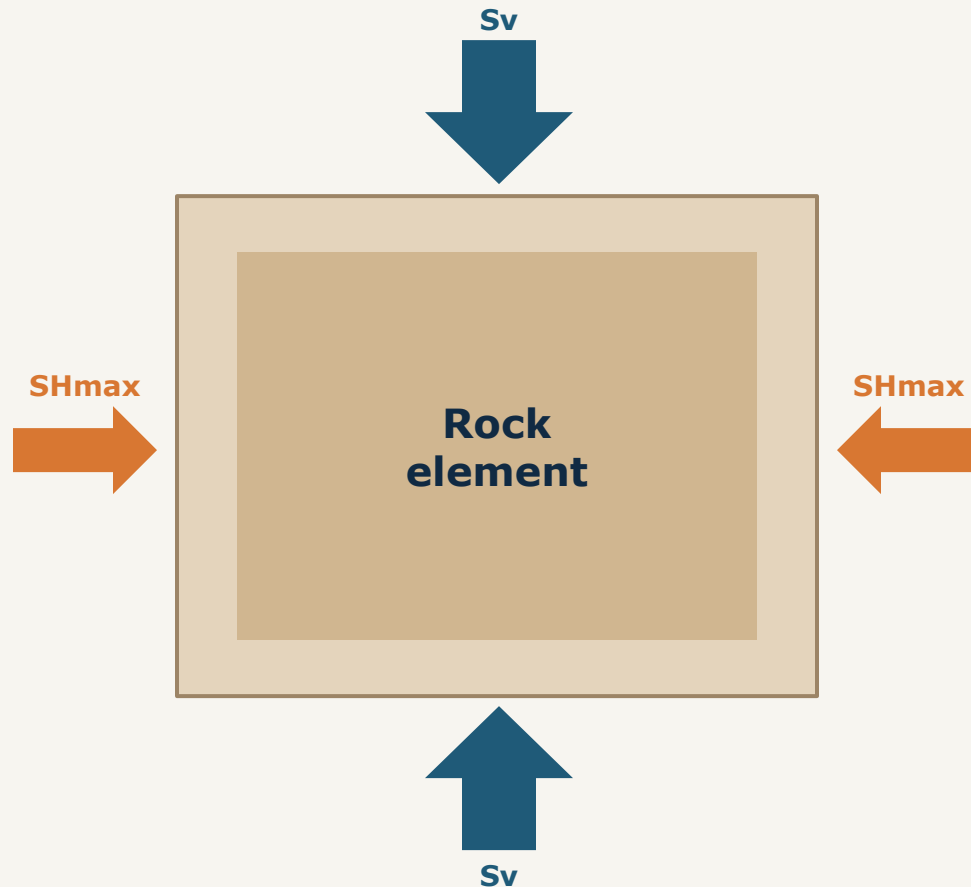
MPa or psi

- Compressive strength
- Tensile strength
- Friction / cohesion

Stress is not strength. Pore pressure is not mud weight.

Stress has magnitude and direction

Principal stresses provide the simplest coordinate system for describing the in-situ stress state



Principal-stress description

- S1** maximum principal stress
- S2** intermediate principal stress
- S3** minimum principal stress

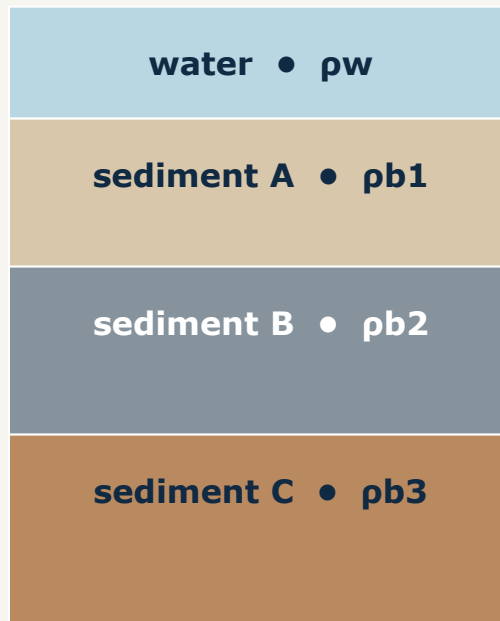
By definition: $S1 \geq S2 \geq S3$

Direction matters because the borehole 'sees' a different stress state when its trajectory changes

Vertical stress accumulates from the weight above

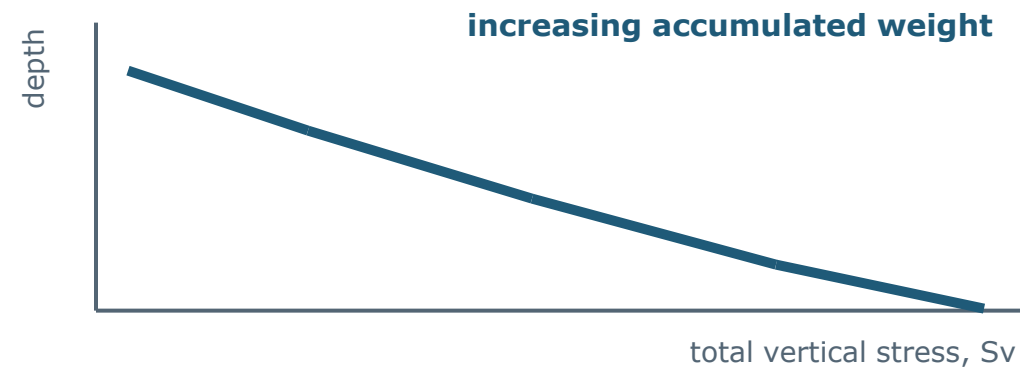
Density integration converts the overburden column into total vertical stress

OVERBURDEN COLUMN



$$S_v(z) = \int \rho_b(z) g dz$$

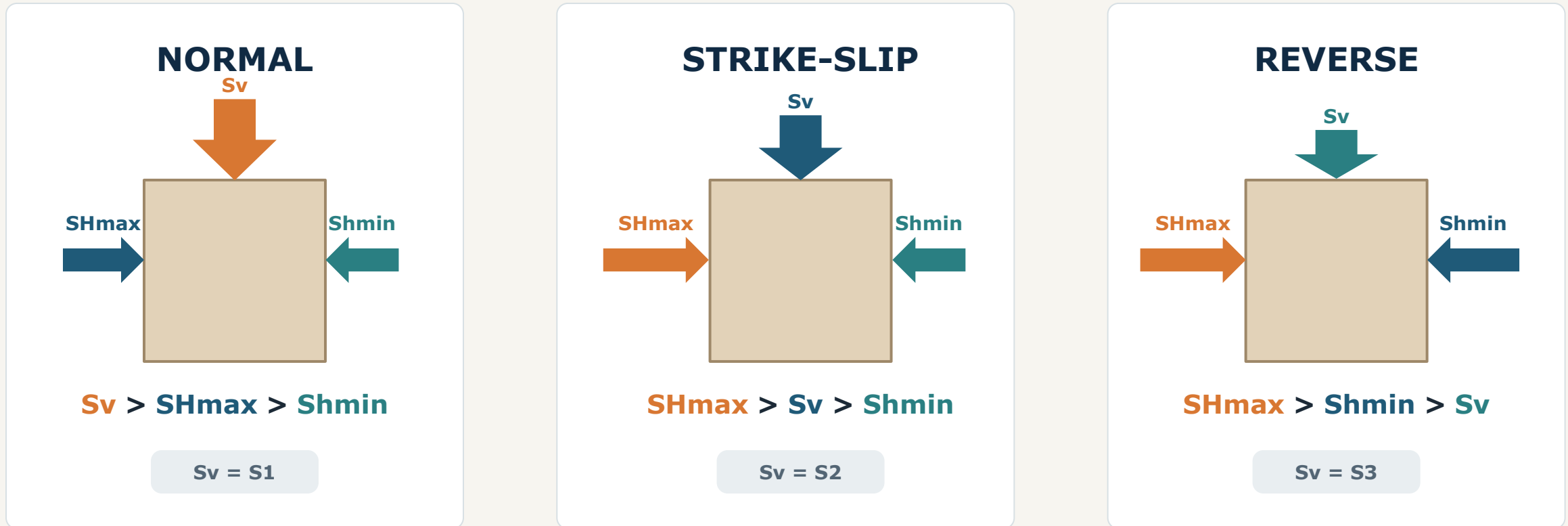
Offshore: include the water-column contribution to total overburden



Density uncertainty propagates into S_v —and then into every stress-based calculation

Which principal stress is vertical defines the Andersonian regime

The ordering—not only the labels—distinguishes normal, strike-slip and reverse faulting regimes

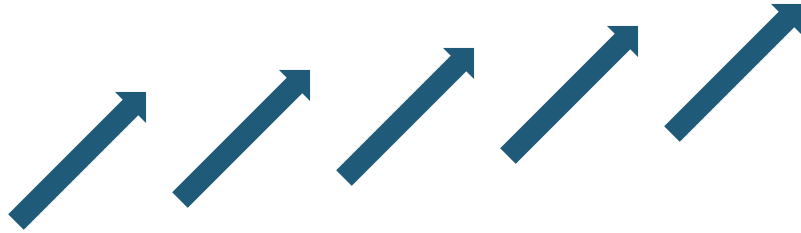


Regime constrains the stress ordering; it does not by itself provide the stress magnitudes

Regional stress is context—not a well-specific measurement

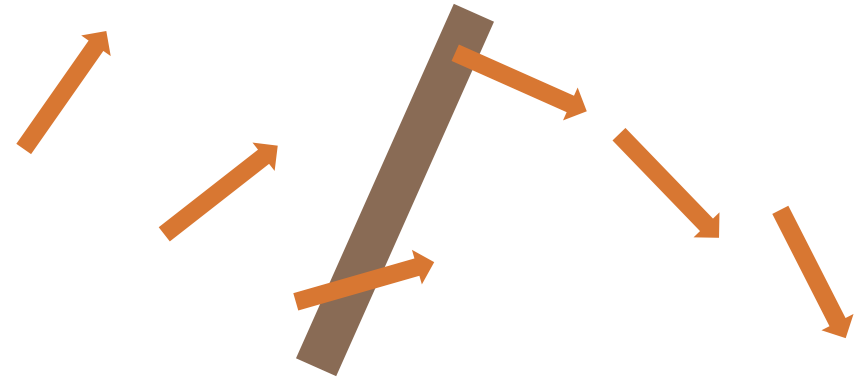
Local geology and data quality can modify or obscure the regional pattern

REGIONAL CONTEXT



Consistent orientations across a broad area can establish a useful first-order trend

LOCAL EVIDENCE

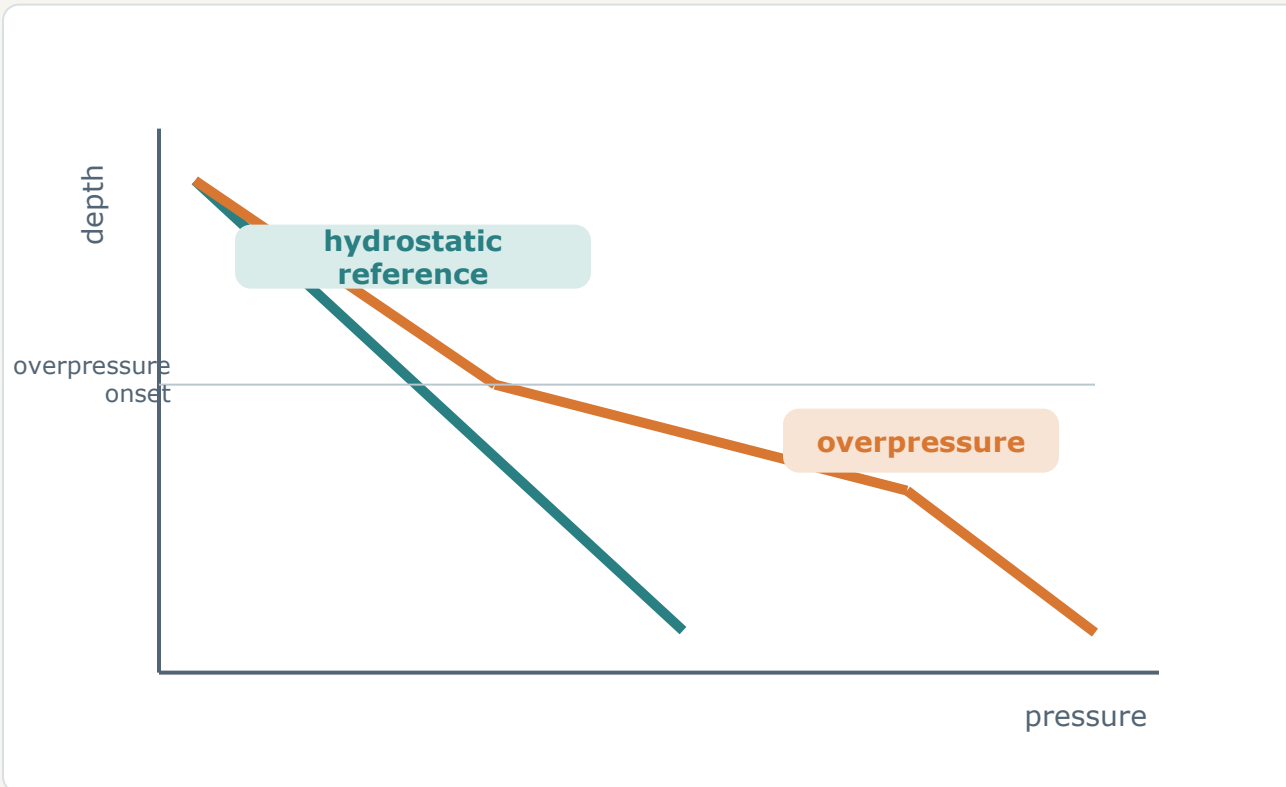


Faults, structural complexity, contrasts and measurement uncertainty can produce local variation

Use regional compilations to set expectations—then calibrate with well-specific indicators

Pore pressure is the fluid pressure within connected pores

It is distinct from overburden stress and from the pressure imposed inside the wellbore



Formation pore pressure

A subsurface fluid-pressure property estimated or measured with uncertainty.

Wellbore pressure

Pressure applied at the borehole wall; affected by MW, ECD, surge and swab.

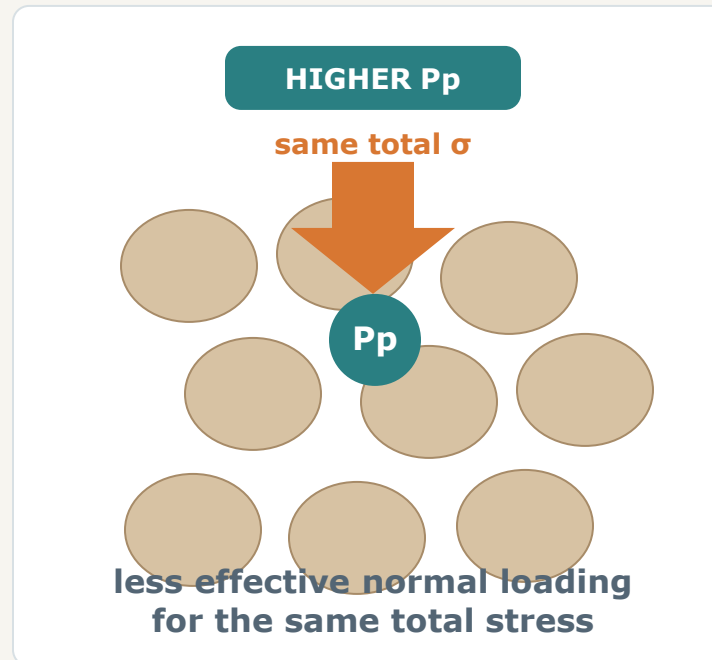
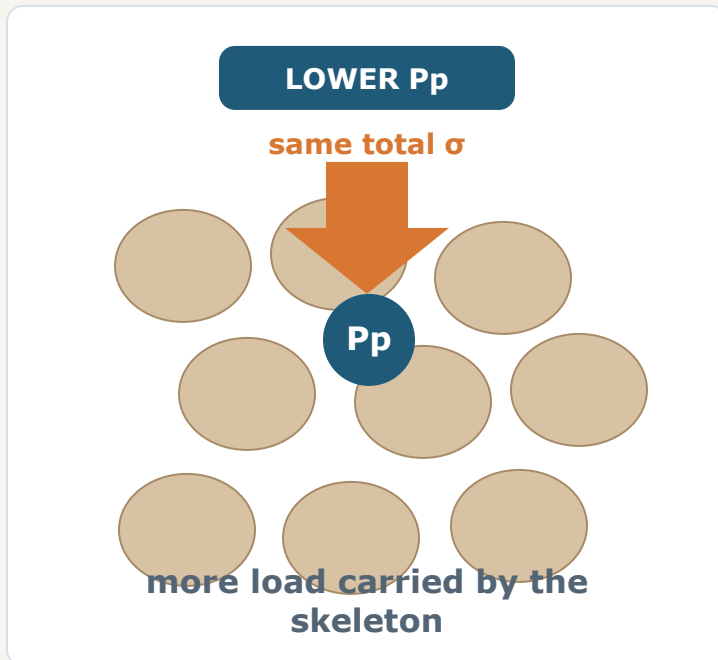
Overburden stress

Total vertical load from the material above—not a fluid pressure.

Pressure predictions should be updated as new well evidence becomes available

Pore pressure changes the stress carried by the rock skeleton

A simple Biot effective-stress form links total stress and pore pressure



Simple normal-stress form

$$\sigma' = \sigma - \alpha P_p$$

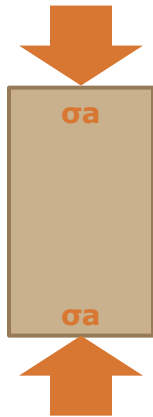
σ' effective stress
 σ total stress
 α Biot coefficient
 P_p pore pressure

Use a consistent sign convention and the appropriate effective-stress formulation

Different tests answer different mechanical questions

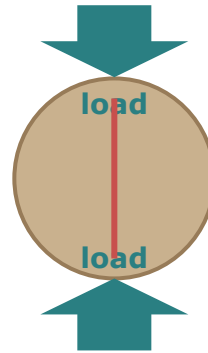
No single laboratory number fully describes wellbore behavior

UCS / uniaxial



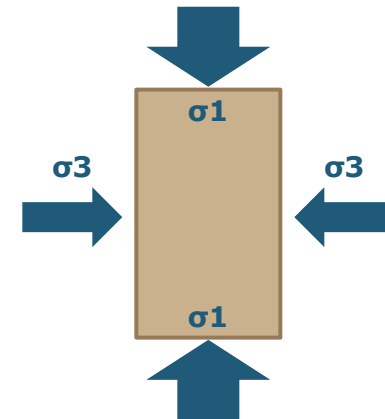
Compressive strength
and stress–strain response

Brazilian / direct tension



Tensile resistance
with method-specific
assumptions

Triaxial



Confinement dependence
and failure-envelope
calibration

**Elastic properties describe deformation; strength parameters describe failure.
They are related—but not interchangeable.**

Rock fabric can make mechanical behavior direction-dependent

Bedding, laminations and fractures challenge an isotropic intact-rock assumption

ORIENTED SAMPLING



normal to
bedding



parallel to
bedding

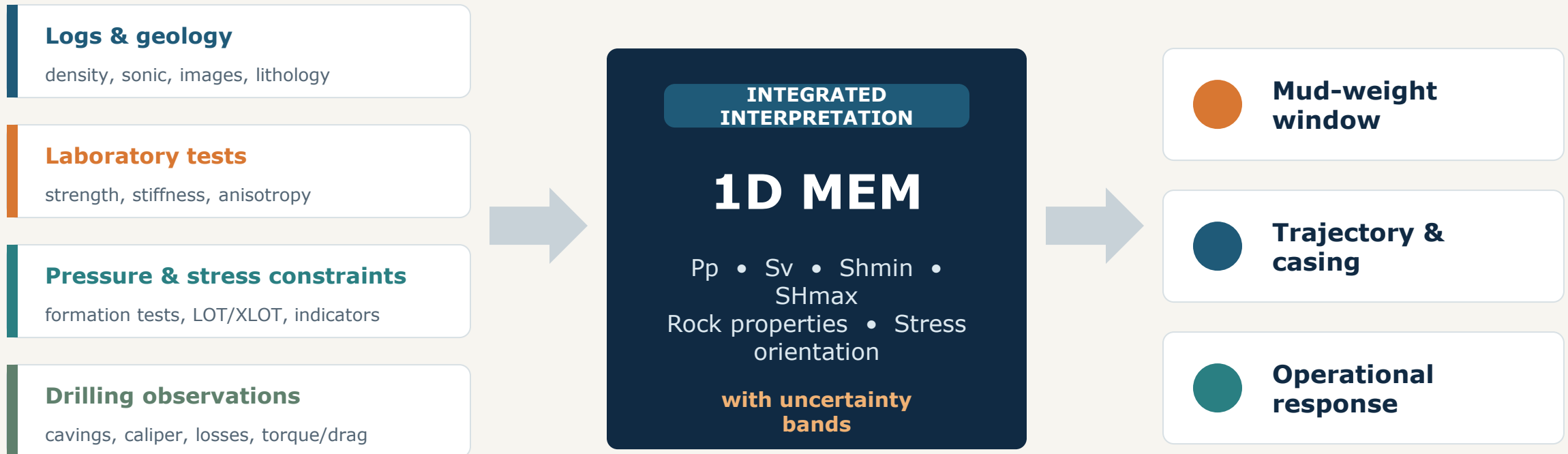
WHAT MAY CHANGE WITH DIRECTION?

-  **Strength** failure stress and failure mode
-  **Stiffness** deformation and stress redistribution
-  **Wave velocity** directional dynamic response
-  **Permeability / pathways** fabric- and fracture-controlled flow

Anisotropy should be characterized—not assumed from appearance alone

A 1D geomechanical model is an evidence-integration workflow

The output should preserve uncertainty and remain testable against drilling observations



No single log measures 'geomechanics'

A useful model states what is measured, what is inferred, and how uncertainty affects the decision

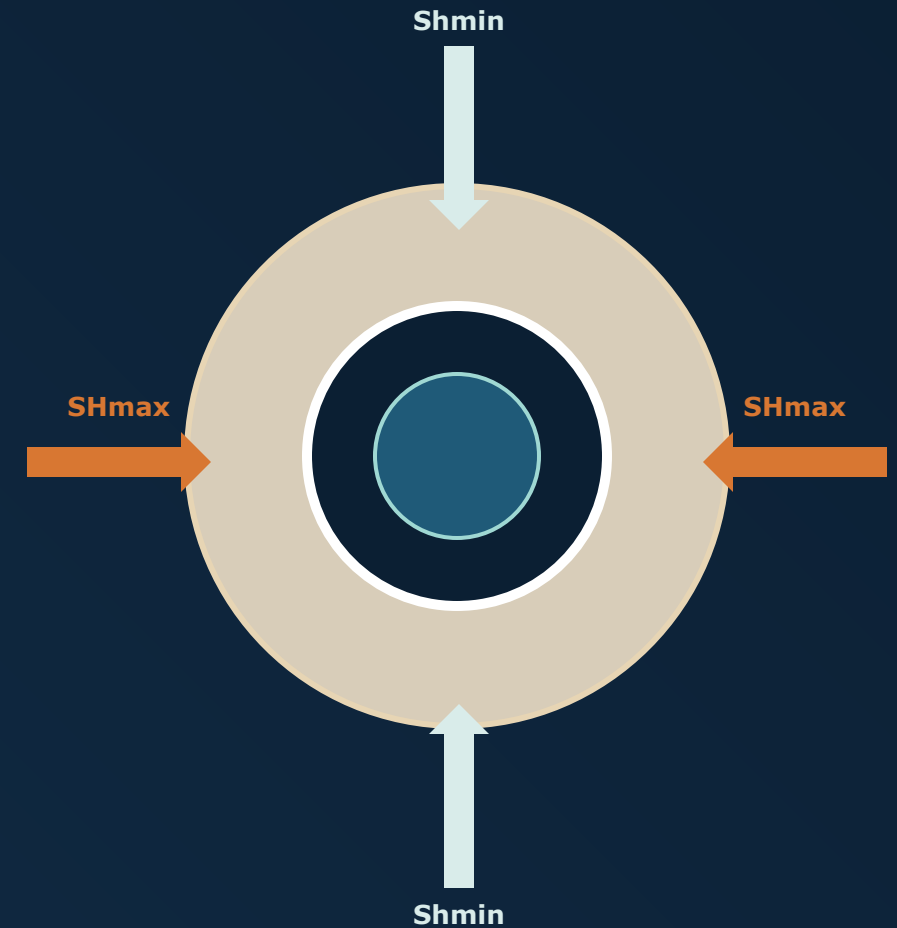
PART II

Then we drill a hole

The far-field model now meets a specific trajectory, mud system and operating history

Focus for this section

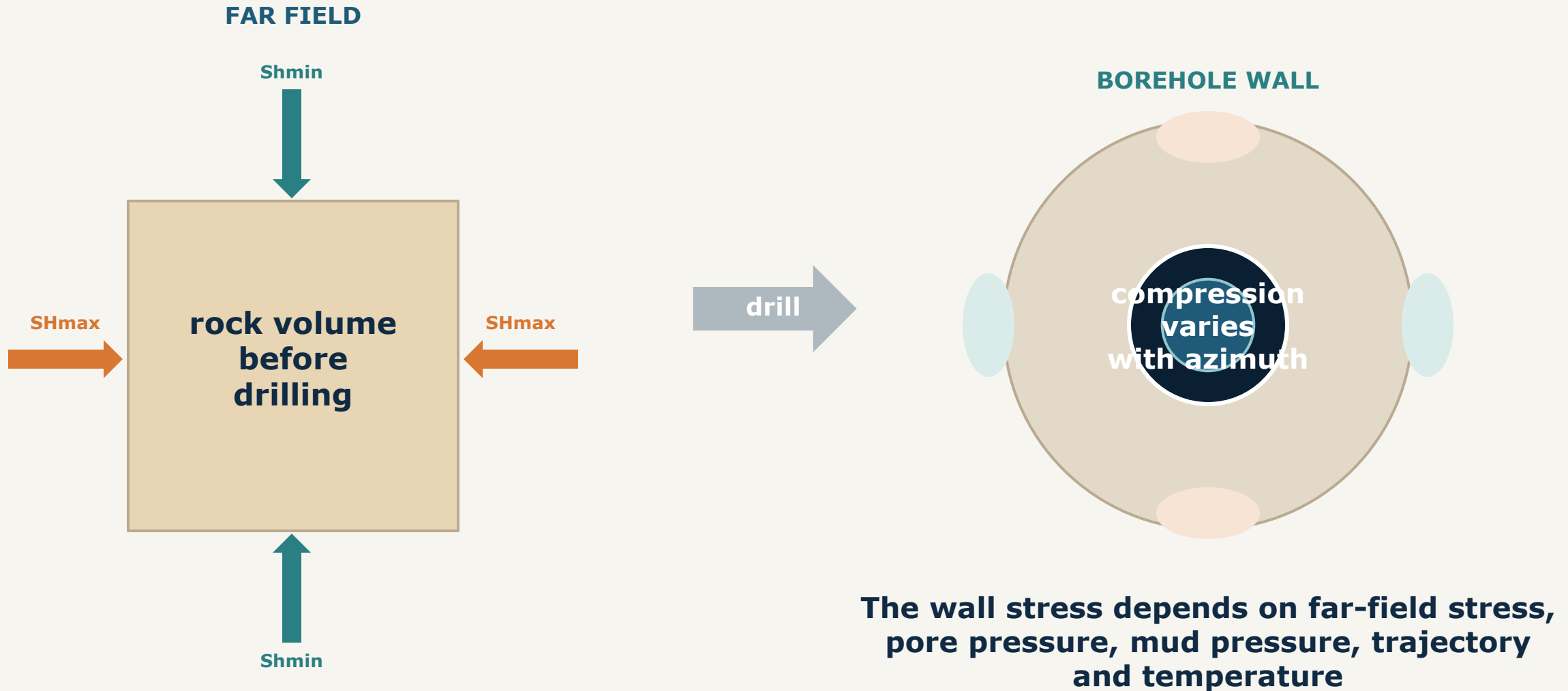
Wall stress, failure evidence, pore pressure and operating window



The borehole changes the stress field around it

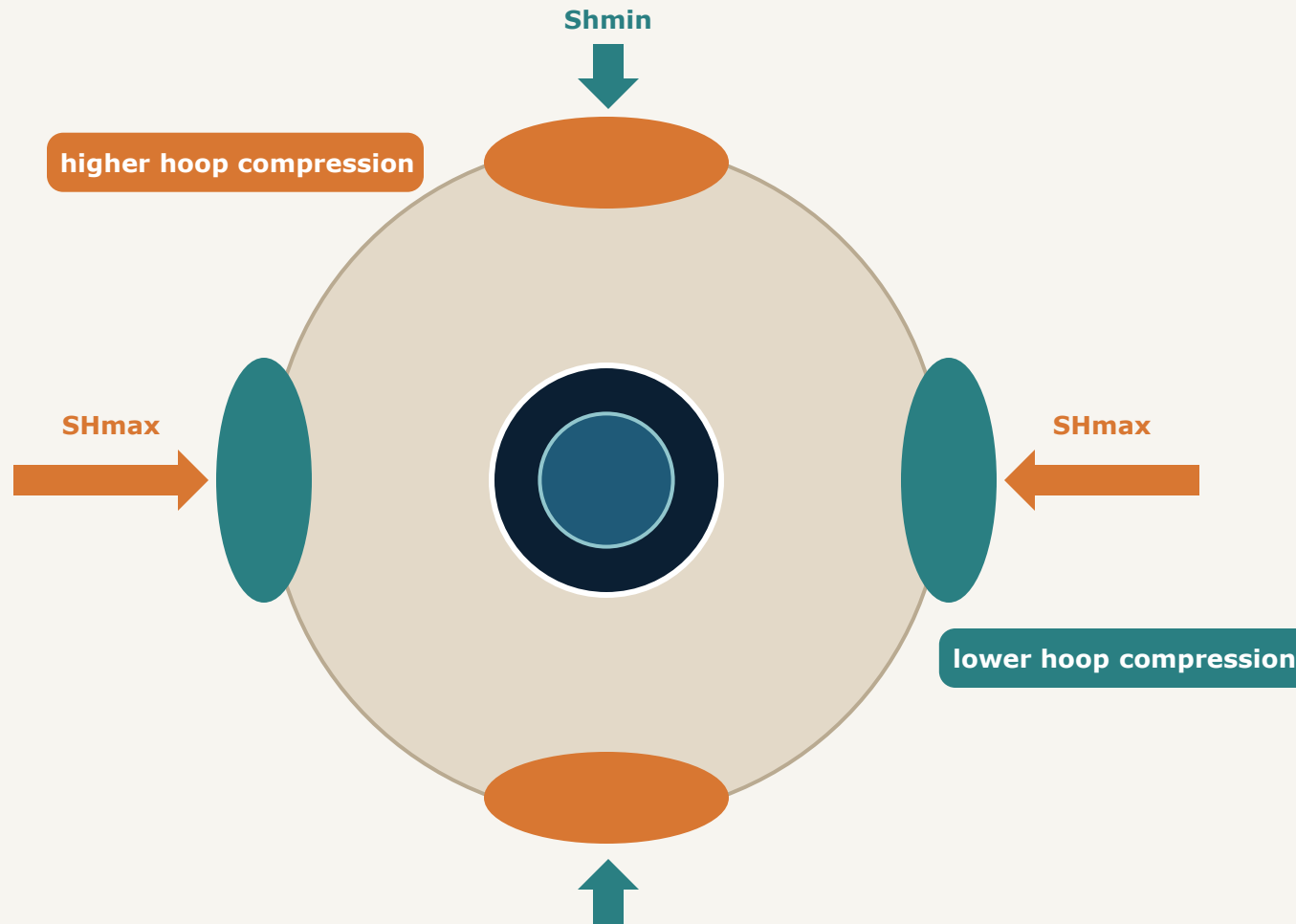
Far-field stress and wall stress are different

Excavating the hole redistributes the pre-existing stress field



Stress concentration around a circular hole

The Kirsch solution provides the basic elastic picture for a vertical borehole



At the wall

$$\sigma_r = P_w$$

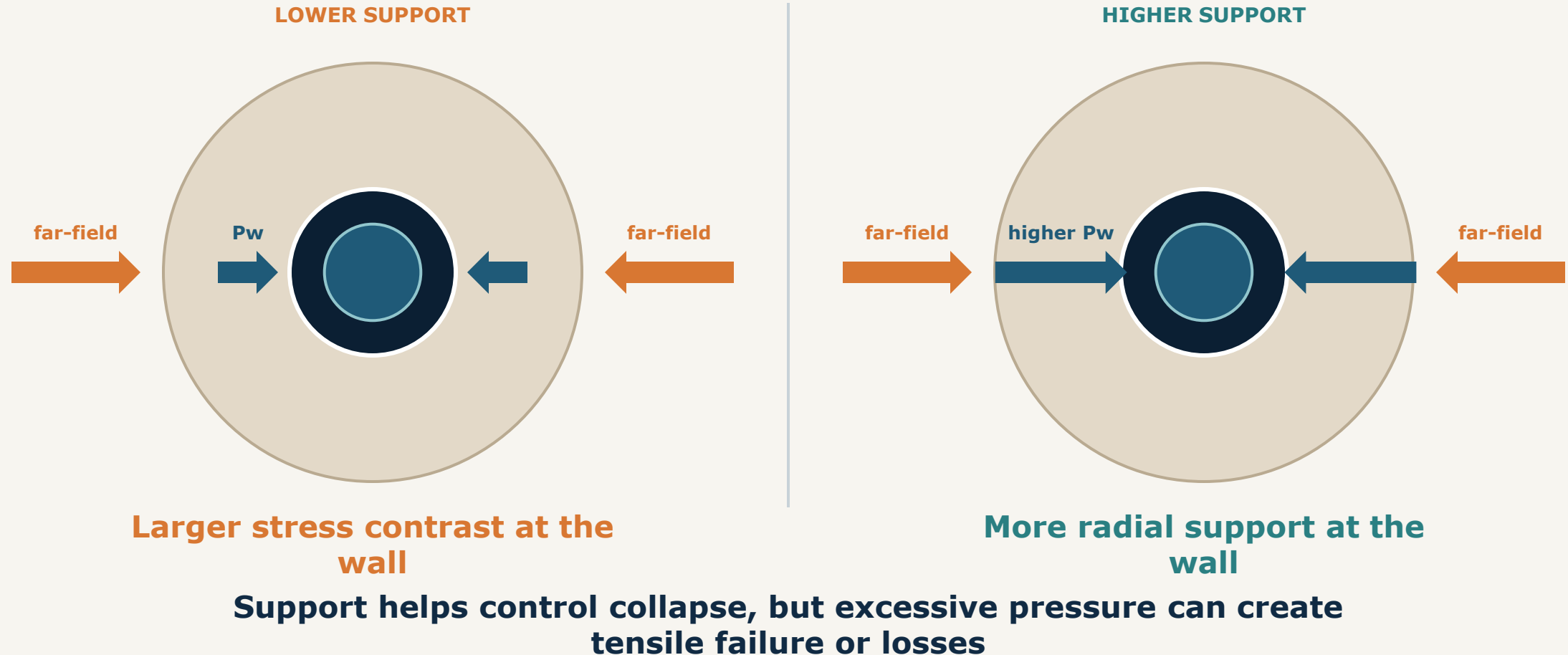
Hoop stress varies with azimuth

Axial stress also changes around inclined holes and under non-isotropic conditions

Conceptual view. Sign convention and angle definition must remain consistent in calculations.

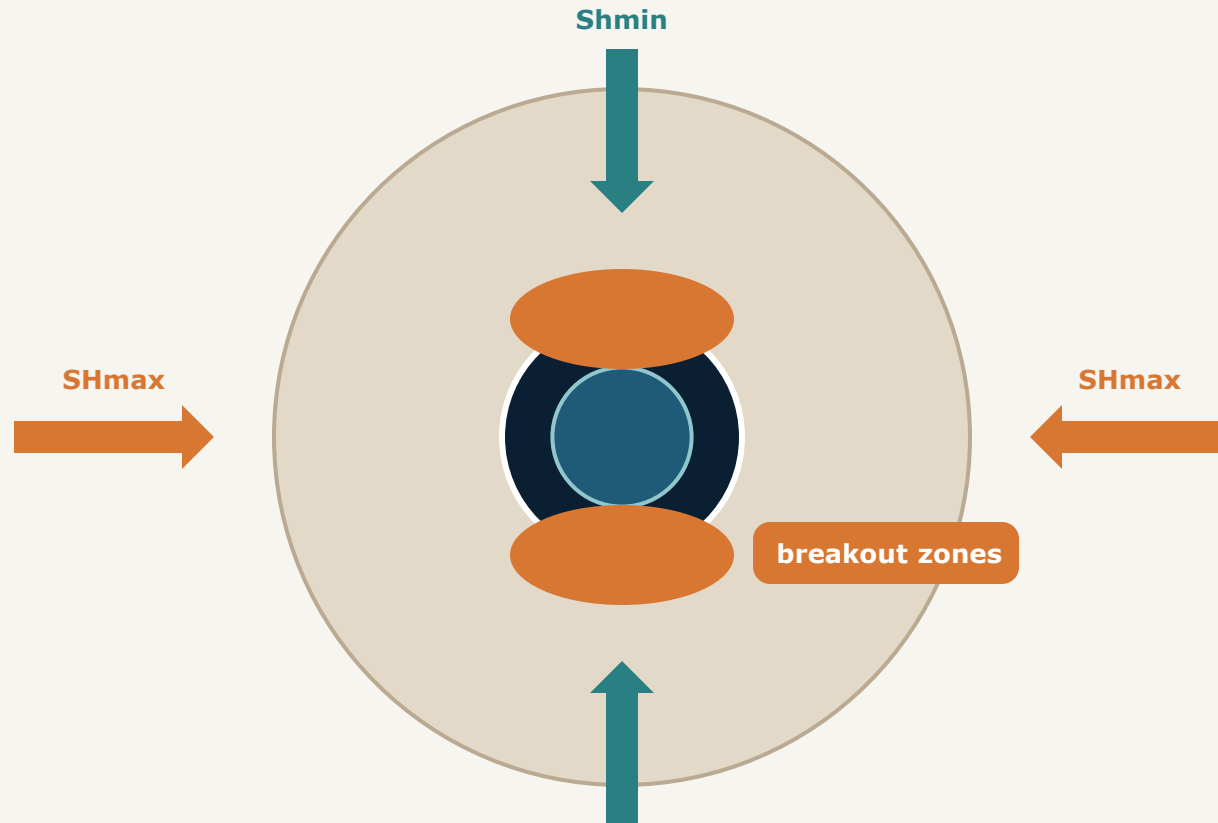
Mud pressure provides mechanical support

Wellbore pressure changes radial stress and the effective wall-stress state



Compressive wall failure produces breakouts and cavings

Failure begins where the effective compressive stress exceeds the available rock strength



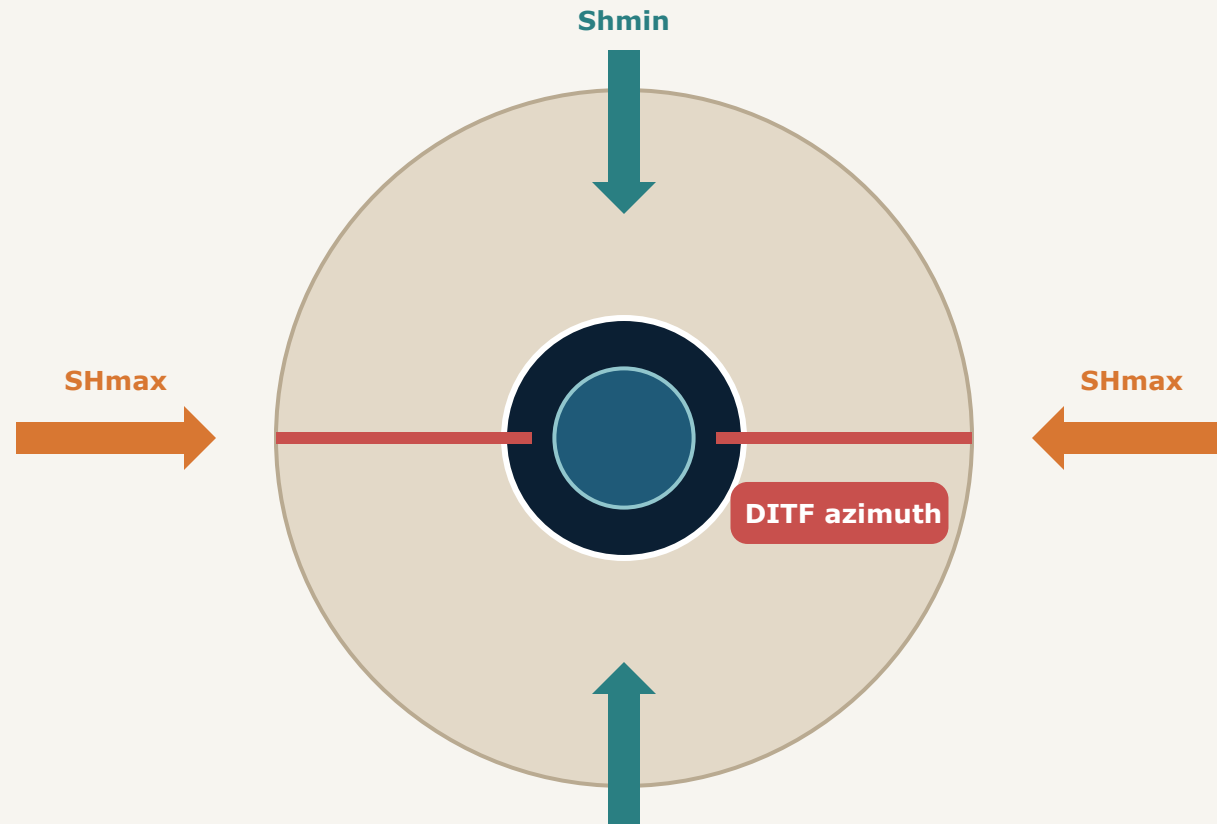
What controls severity?

- Stress magnitudes and trajectory
- Rock strength and anisotropy
- Mud pressure and pore pressure
- Exposure time, chemistry and temperature

Breakout azimuth can constrain stress orientation. Breakout width requires stronger assumptions.

Tensile wall failure creates drilling-induced fractures

Low hoop stress can become tensile under pressure, cooling or an unfavorable trajectory



Possible contributors

- High wellbore pressure
- Cooling-induced tensile stress
- Low minimum wall stress
- Weak planes or pre-existing fractures

A DITF is evidence of tensile wall failure. It does not automatically prove sustained fluid loss.

Breakouts and DITFs provide complementary stress evidence

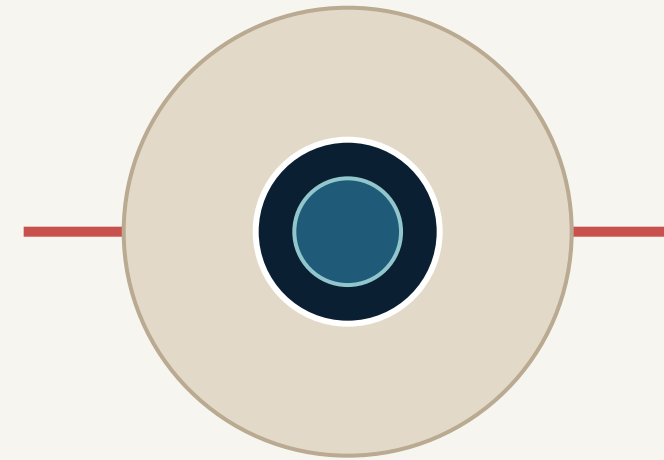
Interpret morphology and azimuth together with trajectory, pressure and rock properties

BREAKOUT



Paired, broad enlargement
Compressive failure
Commonly centered on S_{hmin} azimuth*

DITF

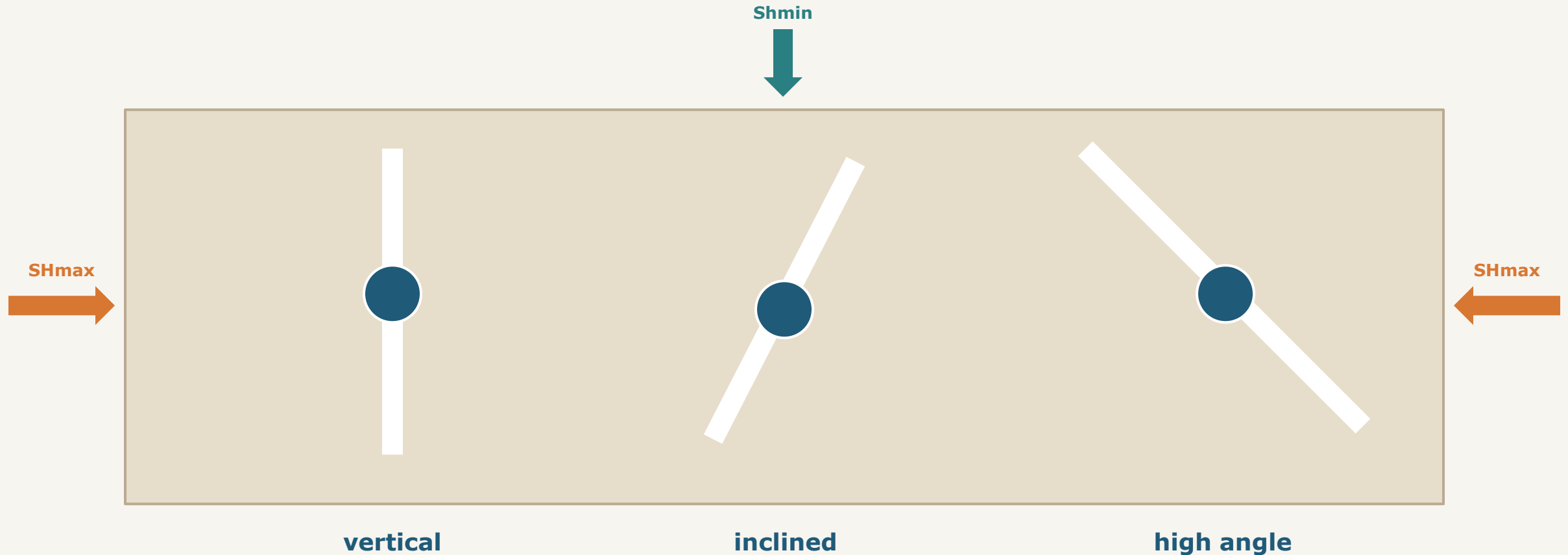


Narrow, sub-parallel traces
Tensile failure
Commonly aligned with S_{Hmax} azimuth*

**Simple vertical well with principal stresses aligned to the borehole. Inclined wells require forward modeling.*

Borehole orientation changes the wall-stress state

The same far-field tensor produces different wall stresses after the trajectory rotates

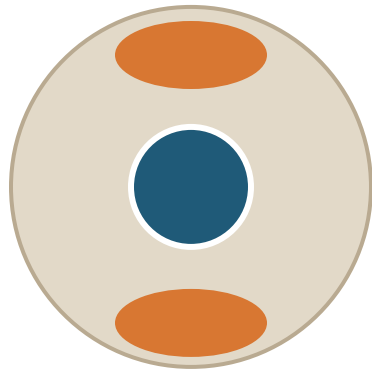


No trajectory is universally safest. The answer depends on the stress regime, azimuth, inclination and rock behavior.

The same formation can behave differently along two trajectories

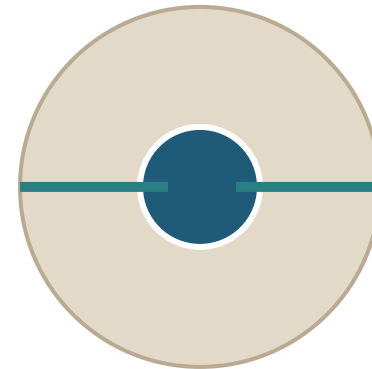
Rock properties and mud weight can stay constant while the wall-stress margin changes

TRAJECTORY A



compressive margin is smaller

TRAJECTORY B



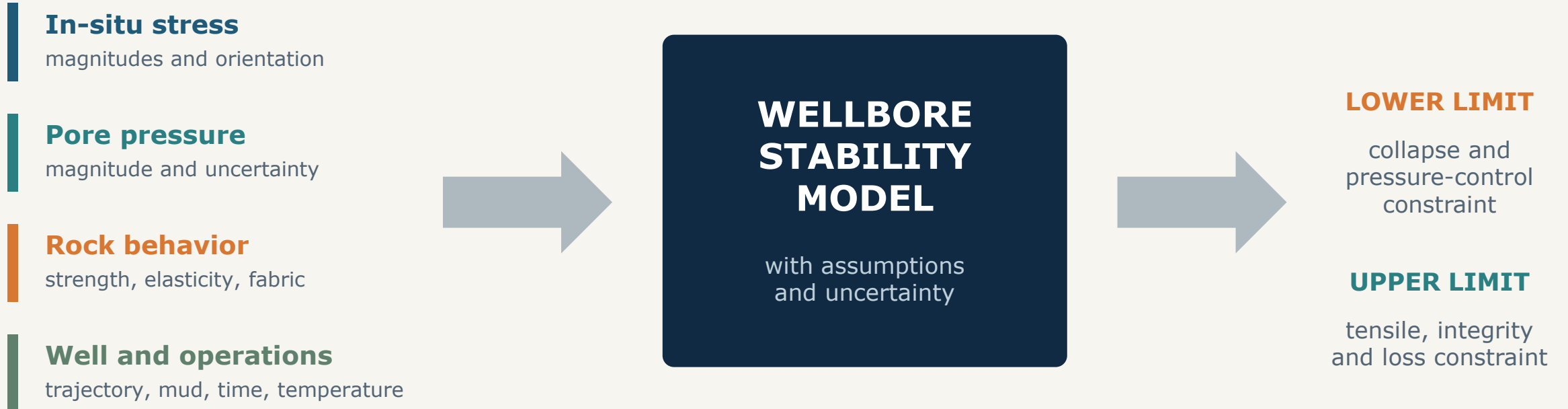
tensile margin is smaller

Same depth, same formation, same mud weight

Different orientation relative to the stress tensor

The central wellbore-stability question

Which wellbore-pressure range keeps failure within an acceptable operational limit?



The useful output is a conditional range, not one exact mud-weight number

Pore pressure reflects fluid support and geological history

Hydrostatic pressure is a reference condition, while several processes can create overpressure or depletion



Hydrostatic reference

Connected fluid column in pressure equilibrium



Disequilibrium compaction

Pore fluid escapes more slowly than burial loads the sediment



Fluid expansion or transfer

Heating, diagenesis, buoyancy or lateral flow can add pressure



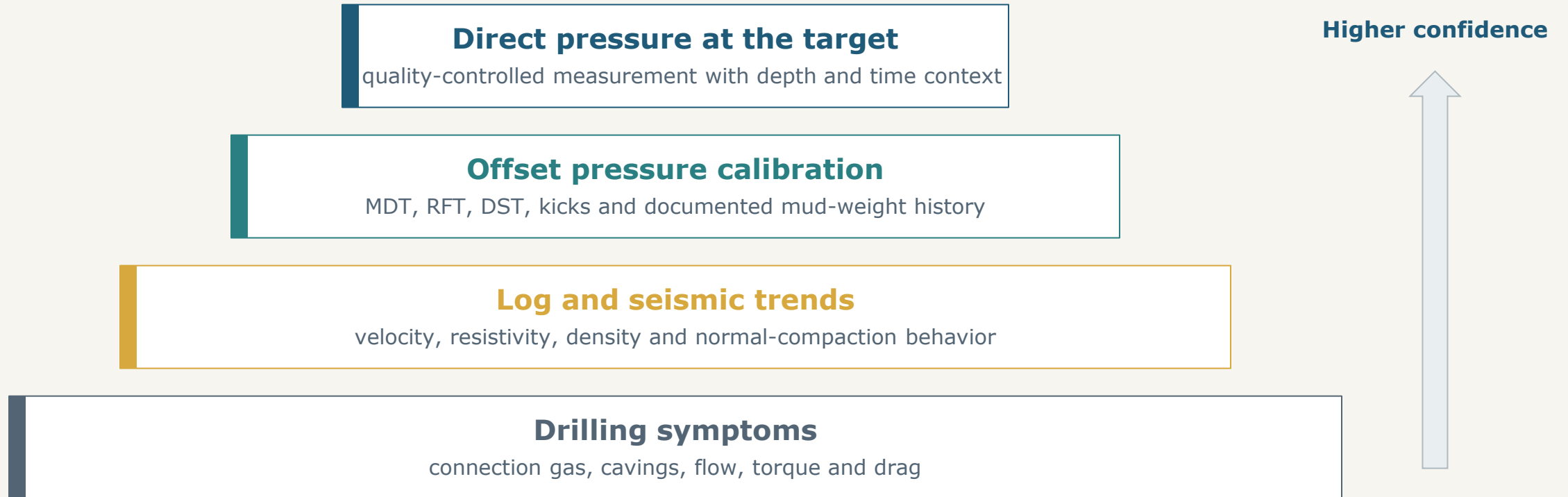
Production or injection

Operations can reduce or increase pressure through time

The mechanism matters because different methods respond differently to unloading and fluid expansion

Pore-pressure confidence depends on the evidence

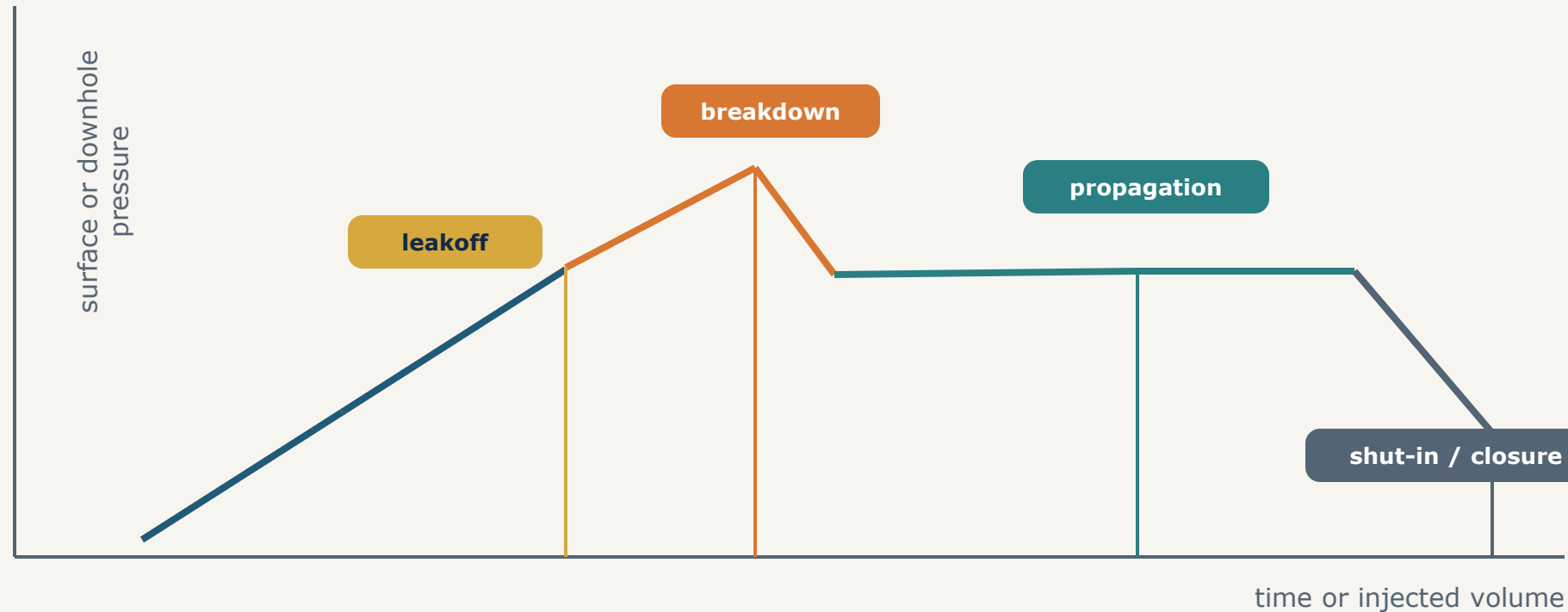
Direct formation-pressure data carry more weight than indirect trends and drilling symptoms



Every pressure model should state its calibration points, blind intervals and uncertainty range

Fracture-gradient terminology can hide different pressures

Test pressure, leakoff, breakdown, propagation and closure describe different parts of the response



A FIT stops below observable leakoff and proves integrity only to the achieved test pressure

FIT, LOT and XLOT answer different questions

Operator procedures vary, so the test name should never replace the pressure-time record

FIT

Primary purpose

Verify integrity to a planned pressure

Typical output

Pass / fail at achieved pressure

Main limitation

Does not measure breakdown or closure

LOT

Primary purpose

Observe deviation from linear pressurization

Typical output

Leakoff pressure under the stated method

Main limitation

Leakoff depends on method and detection rule

XLOT

Primary purpose

Create and cycle a fracture for richer interpretation

Typical output

Breakdown, propagation, shut-in and possible closure information

Main limitation

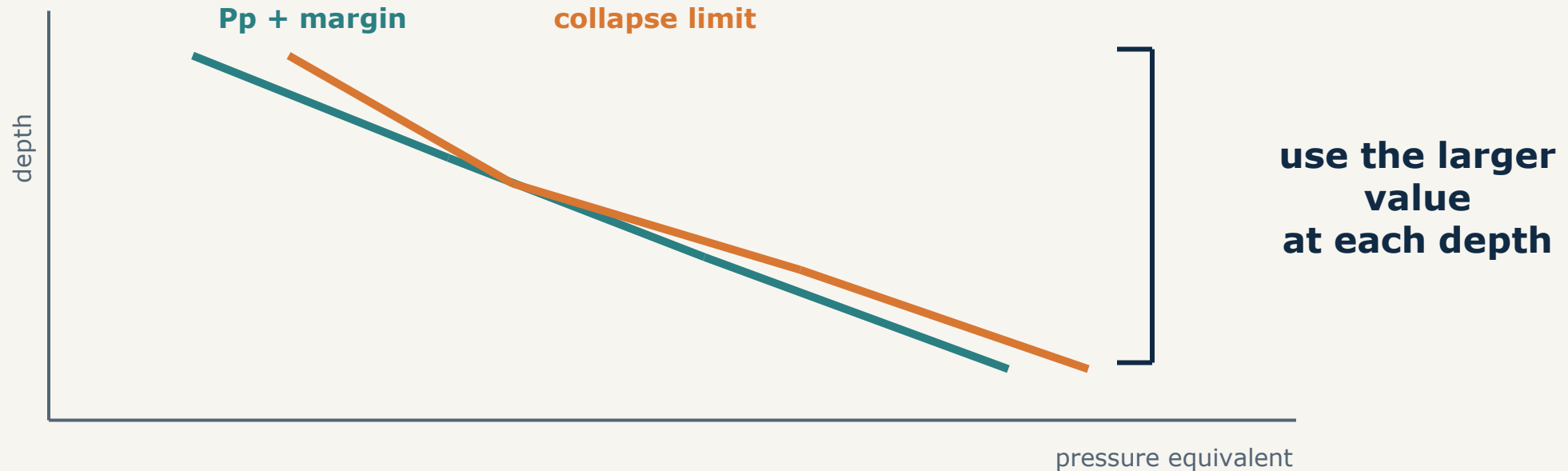
Interpretation needs good rate, volume and pressure data

Record depth reference, fluid density, temperature, pump rate, volume, pressure location and interpretation method

The lower operating limit follows the controlling constraint

Pressure control and compressive stability both set minimum bottomhole pressure

$$P_{\text{lower}} = \max(P_p + \text{margin}, P_{\text{collapse}})$$

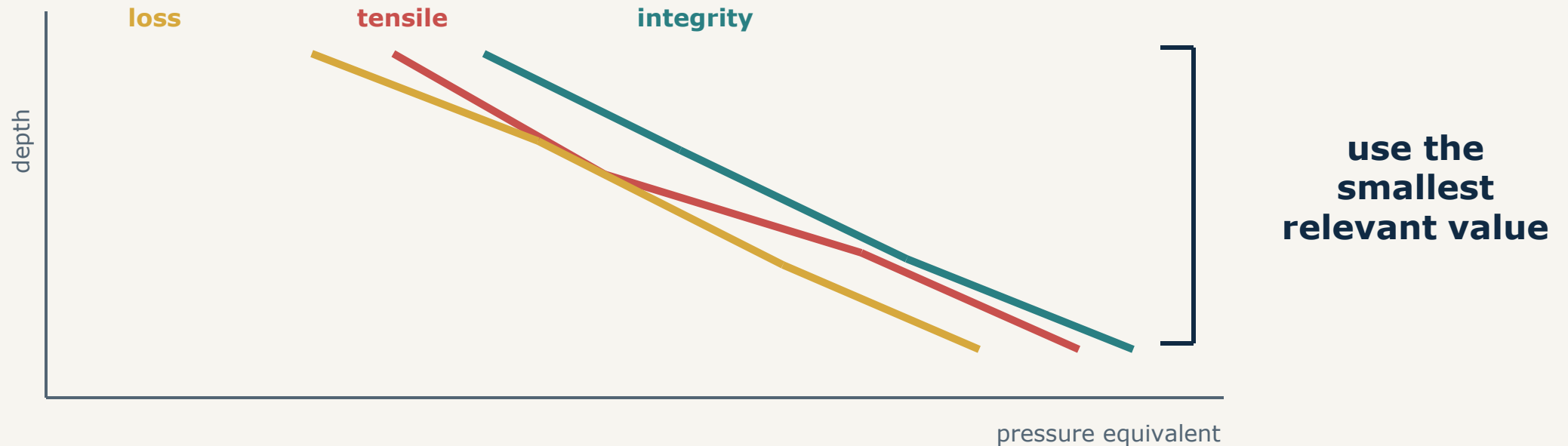


The margin must match kick tolerance, uncertainty and the accepted level of wall damage

The upper operating limit follows the first unacceptable event

Tensile initiation, natural-fracture opening and fluid loss may define different thresholds

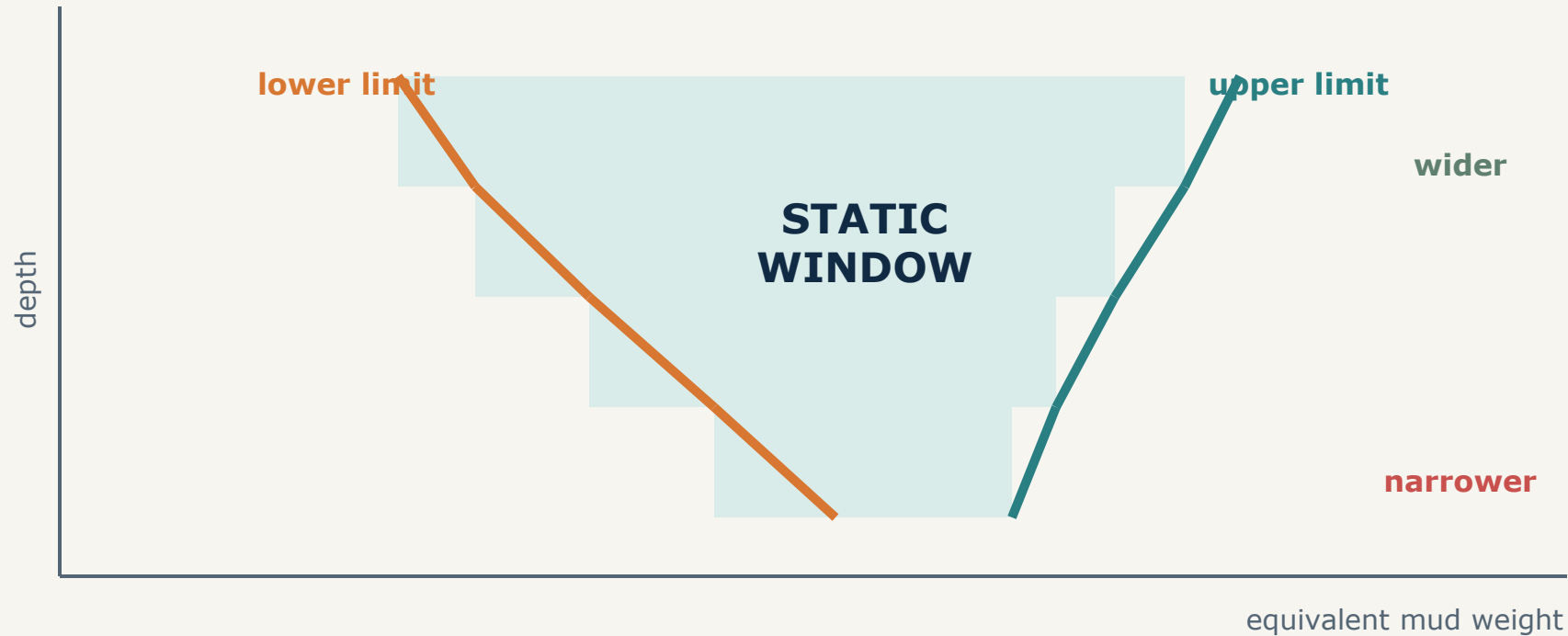
$$P_{\text{upper}} = \min(P_{\text{tensile}}, P_{\text{loss}}, P_{\text{integrity}})$$



The controlling threshold depends on what the operation must avoid and what the available test actually measured

Static mud-window limits

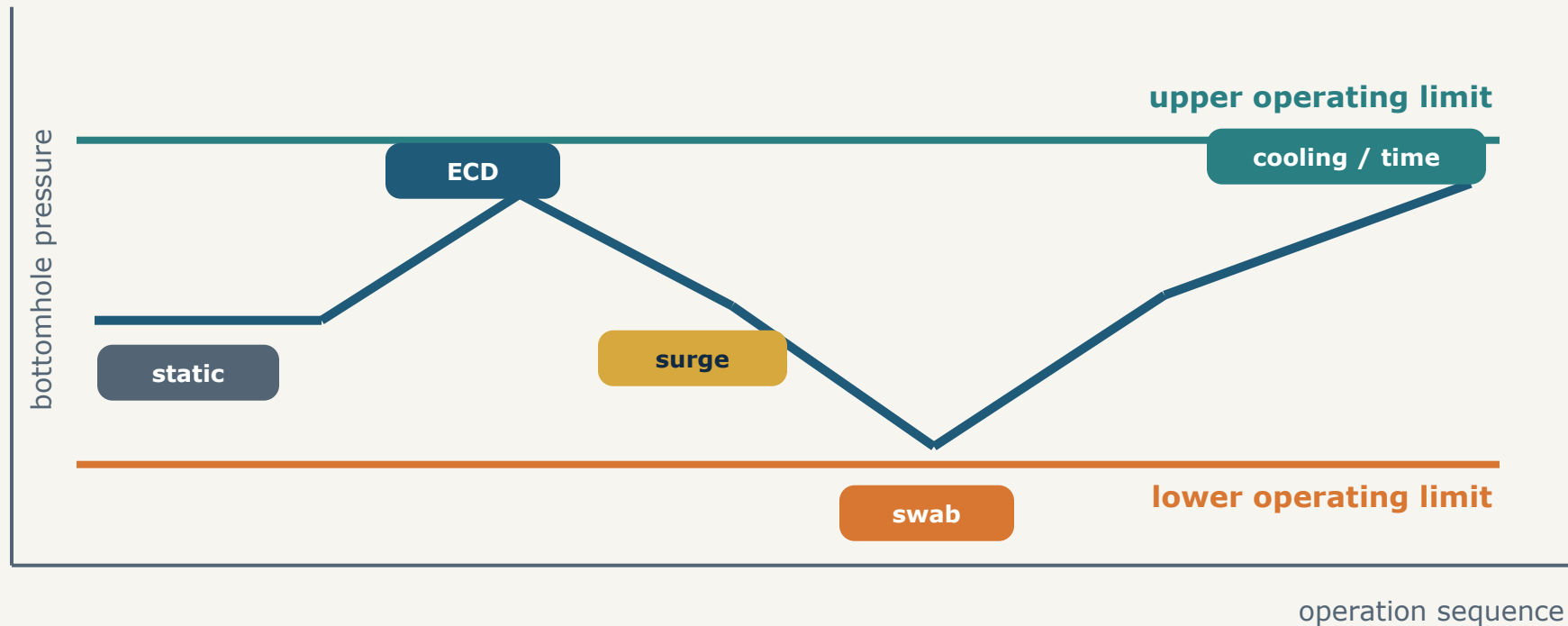
The window changes with depth, trajectory, uncertainty and the selected acceptance criteria



A window can narrow before any single curve crosses. Operational margins may disappear first.

The operating envelope moves during drilling

ECD, surge, swab, temperature and time shift actual bottomhole pressure or the failure limits



OPERATING QUESTION

Where is the actual BHP?

surface MW alone does not answer it

The next decision uses the dynamic bottomhole-pressure envelope, not the static mud-weight curve alone

The well starts talking

Drilling observations reveal when the actual well departs from the pre-drill model

Focus for this section

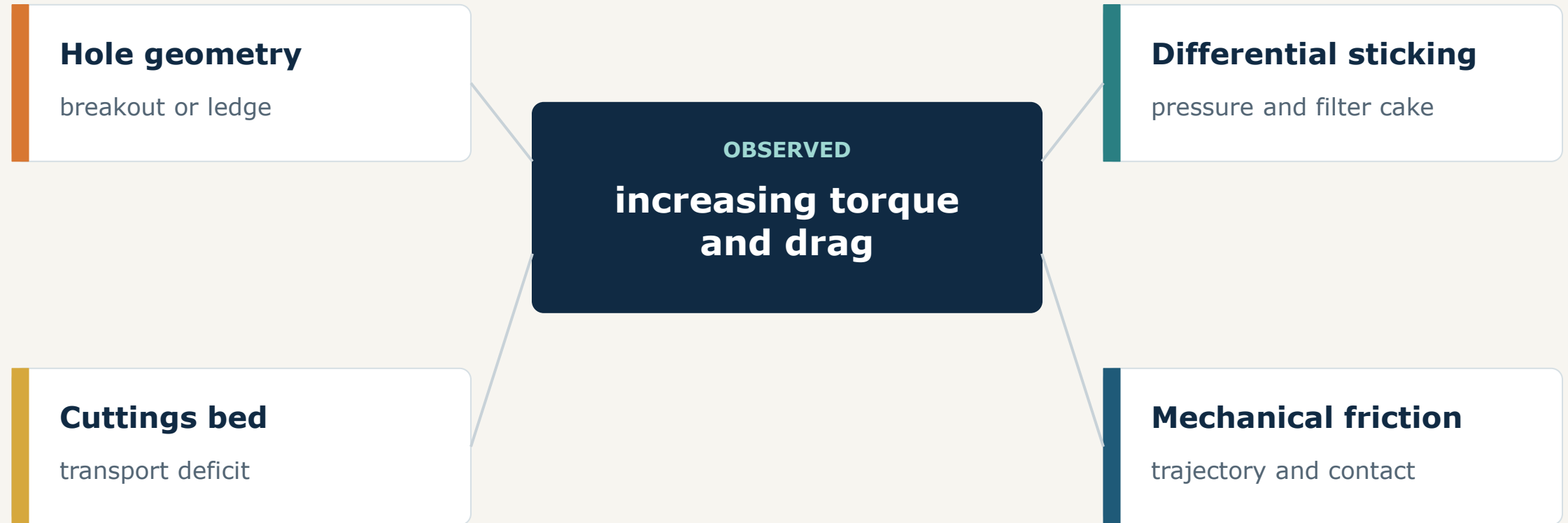
Evidence, diagnosis, response and model update

- **SURFACE**
torque, drag, SPP, returns
- **DOWNHOLE**
ECD, vibration, image, caliper
- **ROCK**
cavings, cuttings, texture
- **TIME**
onset, persistence, recovery

Interpret the pattern, not one indicator

A symptom is evidence, not yet a mechanism

Several mechanisms can produce the same surface observation

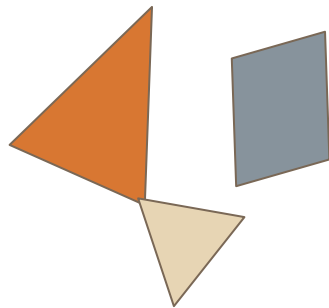


The next question is which mechanism explains the whole evidence pattern

Cavings preserve part of the failure story

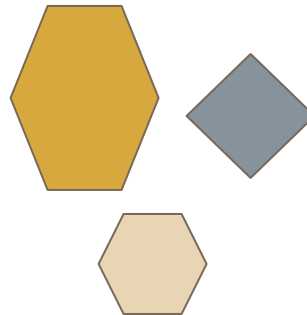
Shape, size, surface texture and recovery timing add useful clues

Angular or splintery



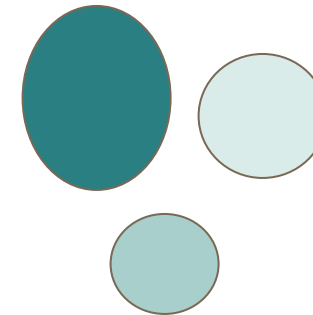
May support a stress-driven failure interpretation

Blocky



May reflect bedding, natural fractures or breakout growth

Soft or dispersed



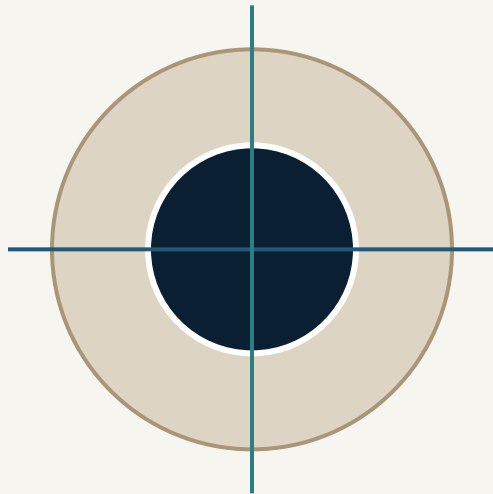
May support a rock-fluid interaction or transport interpretation

Caving morphology is suggestive, not a unique fingerprint

Hole shape adds spatial evidence

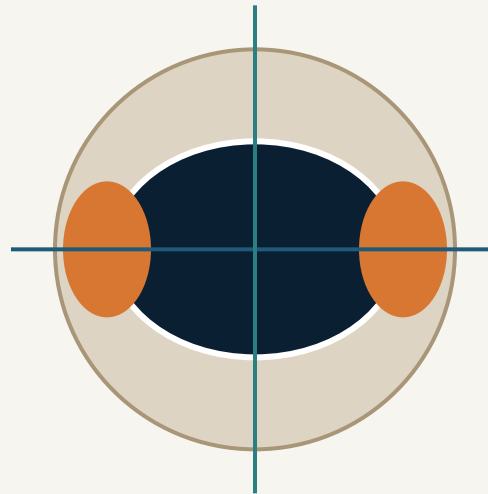
Caliper and image data help separate directional failure from general enlargement

Gauge hole



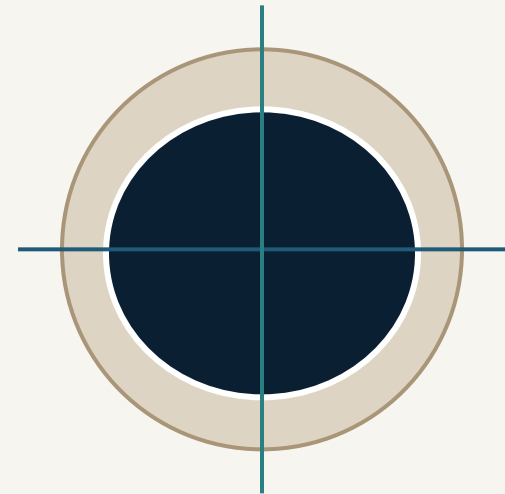
arms remain near bit size

Directional enlargement



opposed enlargement persists in azimuth

Broad washout

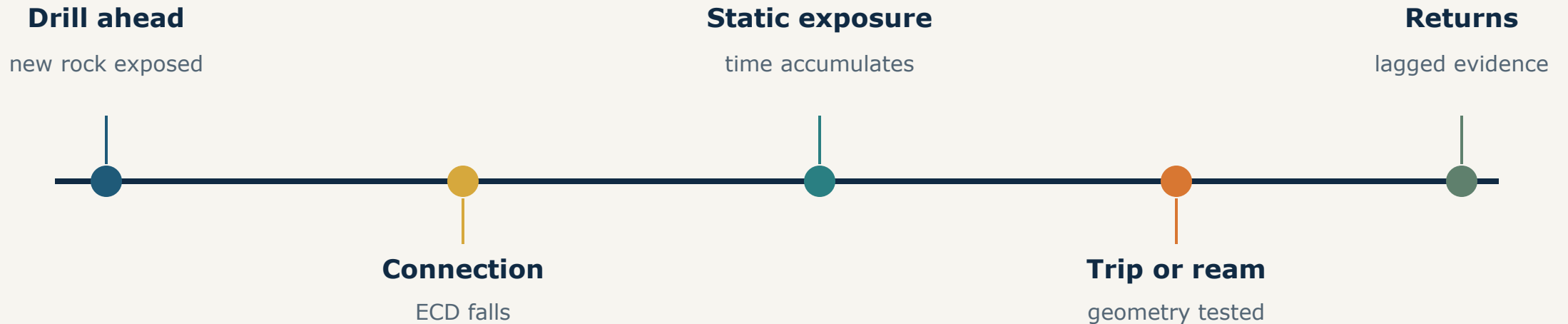


enlargement lacks a stable azimuth

Tool response, trajectory and acquisition conditions still matter

Time links the symptom to the process

Onset and recovery help distinguish immediate pressure effects from delayed weakening



Record the first onset at the source depth, then correct returned solids for lag

Surface trends show when the system changed

Aligned time and depth tracks reveal patterns that isolated values hide



Shared onset strengthens a common-cause interpretation

The strongest diagnosis reconciles independent evidence

Confidence rises when prediction, measurement and recovered rock tell the same story



Contradictions are useful because they expose a weak assumption or missing observation

Mechanisms leave different evidence patterns

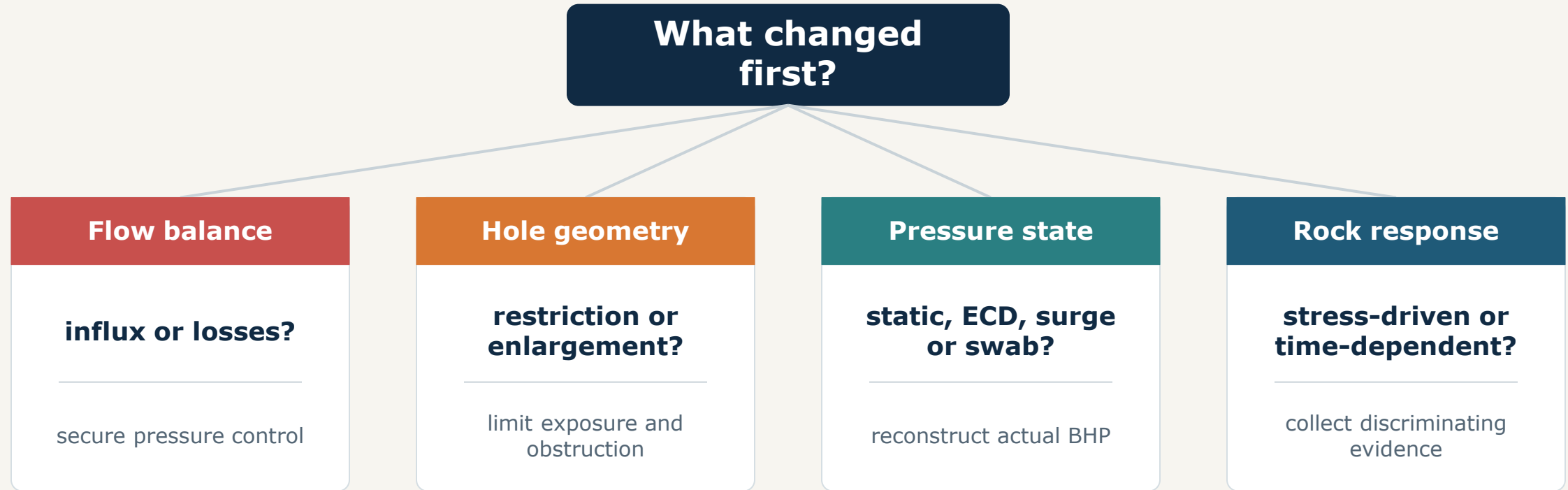
The pattern across observations helps rank competing explanations

Mechanism	Returns	Hole shape	Timing	Pressure response
Compressive failure	angular cavings	directional enlargement	may grow with exposure	often improves with support
Tensile opening or loss	reduced or lost returns	fracture evidence may be subtle	linked to pressure peaks	worsens as BHP rises
Rock-fluid weakening	soft or dispersed solids	broad enlargement possible	often delayed	MW change may not resolve
Hole cleaning deficit	variable cuttings recovery	cuttings bed or pack-off	linked to transport state	responds to hydraulics and movement

Use this as a hypothesis screen, not a universal lookup table

Diagnosis narrows the safe action set

Each branch tests a different immediate risk before changing the drilling program



The first action should protect barriers while preserving diagnostic evidence

Mitigation changes pressure, exposure or rock response

Selected lever must address the diagnosed mechanism and respect the opposite limit

1

Bottomhole pressure

MW, ECD,
backpressure, surge
and swab

2

Exposure

open-hole time, trip
sequence and casing
point

3

Mechanical loading

trajectory, dogleg and
drilling practice

4

Rock-fluid response

activity, inhibition,
sealing and
temperature

Choose the smallest change that tests the diagnosis without consuming the remaining margin

Mini-case: hole condition deteriorates after 10 hours

Rank the mechanism, identify the missing evidence and choose the next safe action

OBSERVATIONS

- Static MW remains within the planned window
- Torque and drag rise after a long connection
- ECD returns to plan while circulating
- Large angular cavings reach surface after lag
- No sustained gain or loss is observed

Depth 3,420 m MD

10 h exposure

01

Most likely mechanism

Which explanation fits every observation?

02

Missing evidence

What would separate the top two hypotheses?

03

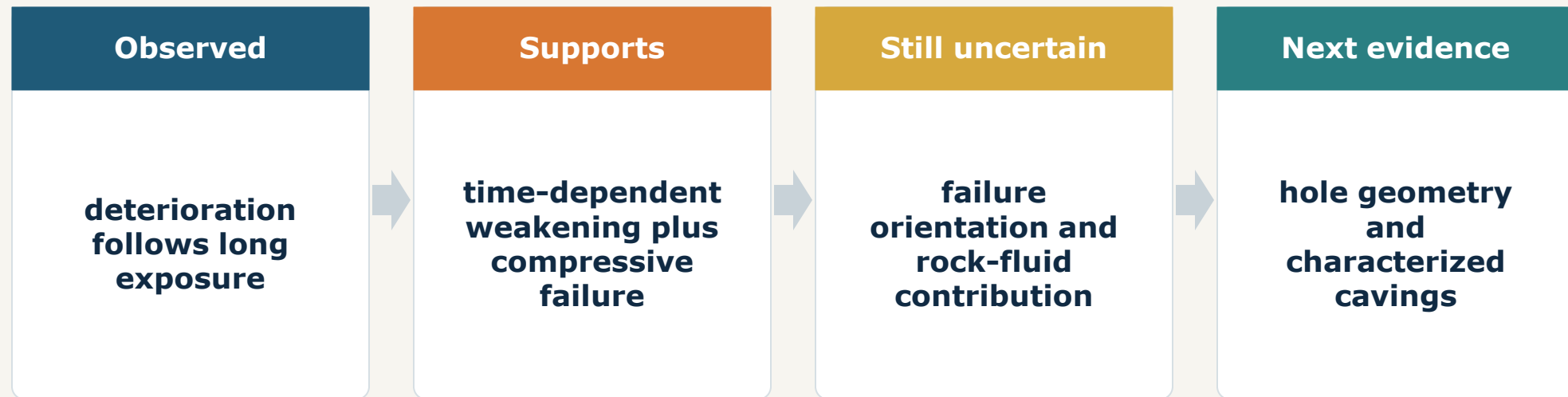
Next safe action

How will it protect both pressure limits?

Discuss in pairs for two minutes

Debrief: the evidence favors delayed shale weakening

The conclusion remains provisional until geometry and rock evidence confirm it



NEXT: AMICS examines mineralogy and microtexture

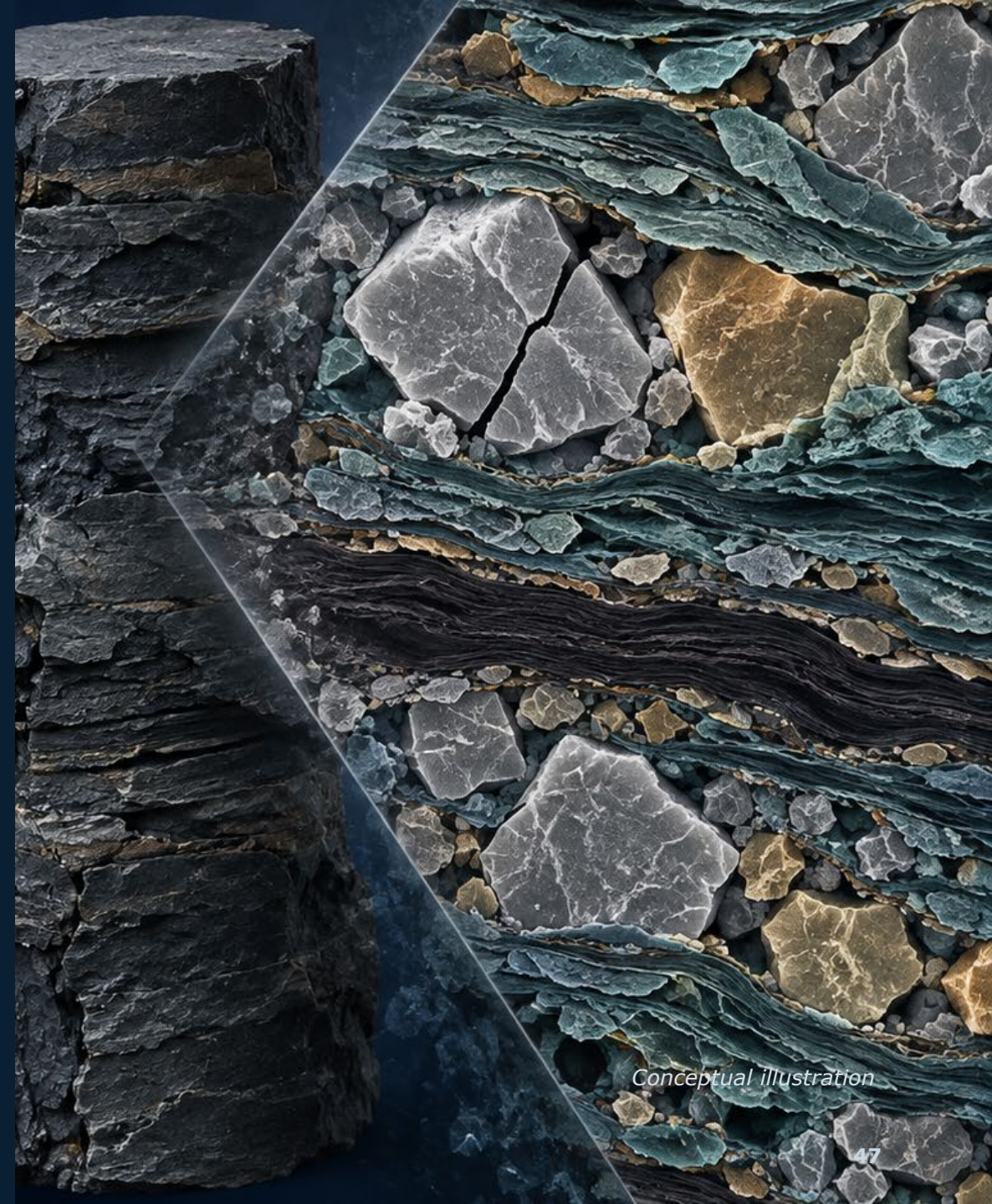
The question shifts from what failed to which fabric and minerals controlled the response

PART IV

Mechanical behavior begins with the rock

Mineralogy, texture and fabric influence how a specimen responds to stress, pressure, fluid and time

AMICS technical context supported by Hitachi High-Tech



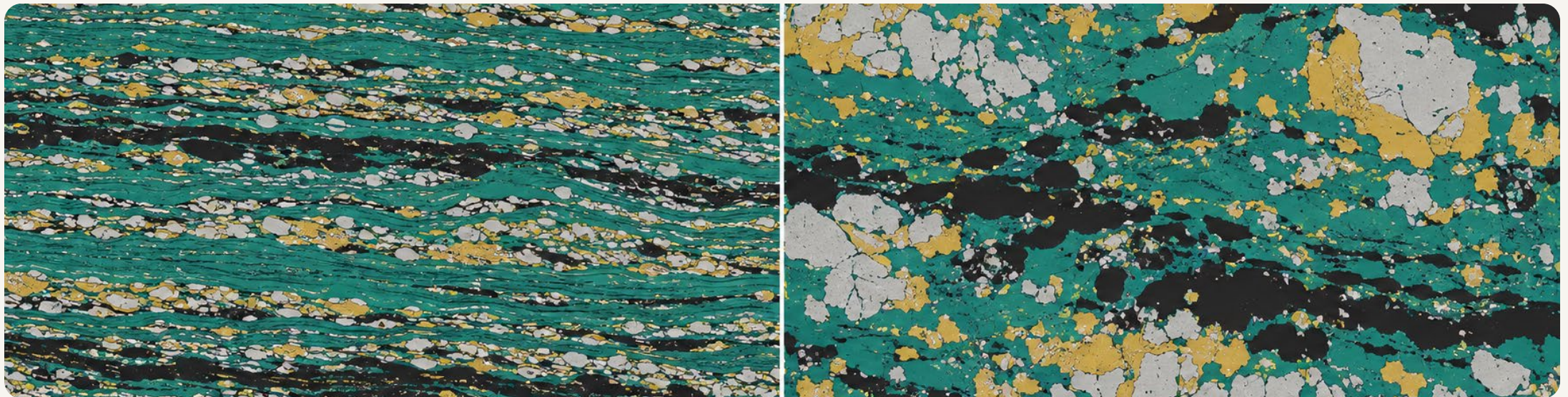
Conceptual illustration

From bulk mineralogy to mineral distribution

Modal abundance alone does not show where phases occur or how they connect

SAME APPARENT BULK FRACTIONS

● clay-rich matrix ● quartz ● carbonate ● organic-rich



Laminated and connected fabric

Clustered and discontinuous fabric

Spatial arrangement can change anisotropy, failure paths and fluid sensitivity

What is AMICS?

Automated phase classification combines electron imaging, EDS spectra and a reference library



PT GEOSERVICES CONTEXT

Hitachi SU3900 with AMICS 5.2

Installed April 2026

AMICS automates phase identification and quantification. The analyst still controls sample preparation, measurement settings, classification rules and geological interpretation.

Where can AMICS add value to geomechanics?

It strengthens material characterization around the tests and models that carry the mechanical decision

USEFUL EVIDENCE

- Mineral and phase distribution across a mapped area
- Lamination, grain association and cement localization
- Specimen representativeness and sources of variability
- Context for weak interfaces and observed failure paths
- Indicators that guide rock-fluid sensitivity testing

REQUIRES OTHER EVIDENCE

Mechanical properties

UCS, tensile strength, elasticity, creep and failure criteria

In-situ state

stress magnitudes, stress orientation and pore pressure

Wellbore limits

collapse pressure, fracture initiation and mud-weight window

Dynamic response

time, temperature, chemistry and operational pressure history

AMICS explains material heterogeneity. Geomechanical tests and models quantify the response.

From microstructure to drilling decision

AMICS contributes one evidence layer within the geomechanics decision loop



ILLUSTRATIVE EVIDENCE CHAIN

- **Material evidence** Clay-rich laminated fabric with localized carbonate cement
- **Well evidence** Angular cavings and deterioration after prolonged exposure
- **Next test** Oriented strength plus fluid-sensitivity and time-dependent testing

NEXT: the integrated case closes the loop from well plan to rig response

The well we have to drill

One planned trajectory. One laminated shale interval. Several plausible geomechanical states.

CASE BRIEF

8 1/2 in	open hole
2,450-3,350 m	measured depth
55° / 125°	inclination / azimuth
1.50-1.55 sg	planned static mud weight

Learning objective: make a defensible drilling decision as new evidence arrives



Synthetic case

Now the subsurface data arrive

The critical shale interval combines stress uncertainty, moderate strength and a directional fabric

REFERENCE POINT

3,100 m MD
2,850 m TVD

Laminated shale within the 8 1/2 in section

AMICS maps phase distribution and helps characterize fabric. Oriented tests quantify the response.

STRESS & PRESSURE

66 MPa

vertical stress, S_v

35 MPa

pore pressure, P_p

48 ± 2 MPa **59-69 MPa**

minimum horizontal
stress, S_{hmin}

retained S_{Hmax}
range

S_{Hmax} azimuth $035^\circ \pm 10^\circ$

INPUT STATUS



constrained



retained range



needs calibration

STRENGTH & FABRIC

UCS

28-44 MPa

**Friction
angle**

24-30°

Bedding dip

12-18°

Fabric

clay-rich
laminae

**Time
response**

poorly
constrained

Construct the geomechanical picture

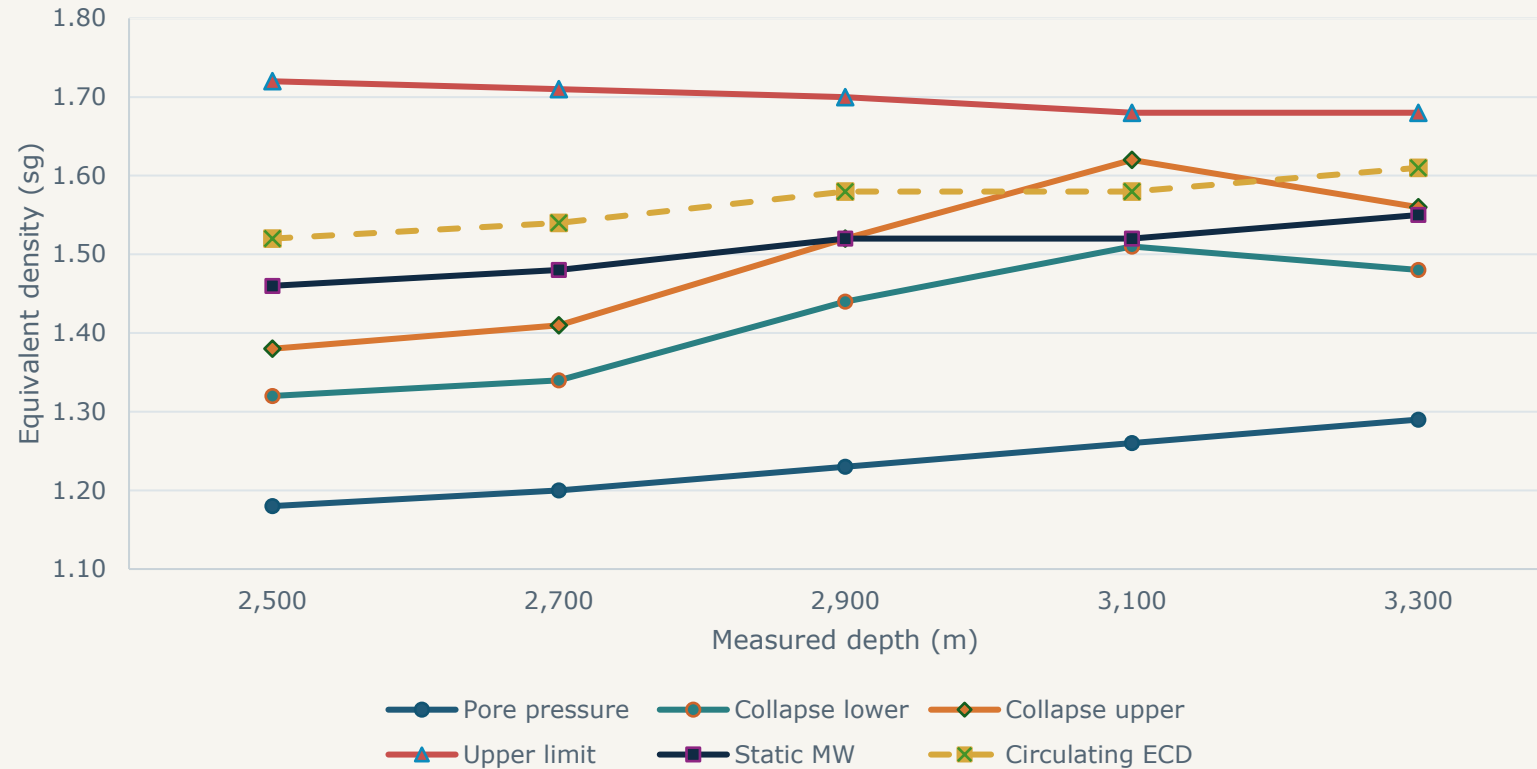
The MEM carries uncertainty forward and exposes which observations can narrow it



The MEM is an evidence ledger: every retained range should point to a measurement, assumption or calibration test

Predict the wellbore response

Trajectory and retained uncertainty convert the MEM into a testable pressure-window prediction



CRITICAL INTERVAL
2,900-3,150 m MD

1.51-1.62 sg retained collapse range

1.52 sg planned static mud weight

1.58 sg circulating ECD

~1.68 sg conservative upper limit

Testable prediction
Instability should worsen after static exposure.
Circulation may temporarily mask it.

Reality arrives at the rig

The sequence of observations matters more than any single symptom

Drill ahead

01

2,900 m MD

Static MW 1.52 sg
ECD 1.58 sg
Torque remains steady

Connection

02

+25 min static

Pickup weight rises
Tight spot develops
No gain or loss

Resume circulation

03

Returns at shakers

Angular cavings appear
Torque improves
Flow remains stable

Repeat exposure

04

Below 3,050 m MD

Pattern recurs
Caliper enlarges
Cavings volume increases

CONSISTENT WITH THE PREDICTION

The problem strengthens after static exposure and eases when circulation adds annular pressure and improves hole cleaning.

ALTERNATIVES STILL TO CHECK

Cuttings transport, ledges, shale-fluid interaction and tool effects can produce overlapping symptoms. Reconcile them with cavings morphology, pressure history and caliper or image data.

Close the loop

Update the model, change the operating plan and define what would disprove the diagnosis

MODEL UPDATE

Raise the lower collapse estimate in the laminated shale from 1.51 to 1.56 sg.

Retain Pp, Shmin and the upper limit. Increase the weight given to fabric and time-dependent response.

CURRENT-WELL RESPONSE

1.58 sg revised static MW

1.61-1.64 sg managed ECD range

Shorter static time connections and trips

Active checks losses, cavings, torque and caliper

ILLUSTRATIVE OUTCOME

Over the next 250 m

- Cavings volume decreases
- Torque trend returns toward baseline
- No losses observed
- Continue monitoring the narrow margin

Decision rule: continue only while the lower-limit margin improves without consuming the loss margin

NEXT: closing principles and Q&A

Seven Things I Want You to Remember

01 **Geomechanics links evidence to drilling decisions**

Rock, stress, pressure and trajectory must be interpreted together.

02 **Pore pressure is only one boundary**

Collapse, fracture and loss limits respond to different mechanisms.

03 **The operating window moves**

Trajectory, ECD, surge, swab and exposure time change the margin.

04 **Rock fabric can control the lower limit**

Anisotropy and shale-fluid interaction may dominate intact strength.

05 **Symptoms need a diagnosis**

Cavings, caliper, torque and pressure history become useful as a pattern.

06 **A model must carry uncertainty**

Retain ranges, state assumptions and define the observation that can narrow them.

07 **The loop closes at the rig**

Update the model and the operating plan when new evidence arrives.



Closing synthesis

Questions / Discussion

Bring the decision you need to make.
We can work backward to the evidence that matters.

Eril S. Lanin, Ph.D.
PT FERG Geosains Indonesia



Jakarta Drilling Society
26 September 2026



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Sign convention and units

Every equation inherits its stress sign, angle definition and pressure reference

Quantity	Convention used here	Typical units
Normal stress	Compression positive	MPa or psi
Shear stress	Sign follows the stated coordinate system	MPa or psi
Pressure	Positive scalar pressure	MPa or psi
Azimuth	Clockwise from geographic north	degrees
Inclination	0° vertical, 90° horizontal	degrees
Density equivalent	Referenced to true vertical depth	sg or ppg

MINIMUM CALCULATION HEADER

- 01 Stress sign and ordering
- 02 Coordinate axes and handedness
- 03 Angle origin and direction
- 04 TVD, MD and pressure datum
- 05 Total or effective stress
- 06 Static MW, ECD or transient BHP

1 MPa = 145.038 psi | 1 sg = 8.345 ppg | 1 sg gives 9.807 MPa/km

Stress tensor and coordinate transformation

The same stress state has different components after the axes rotate

GLOBAL STRESS TENSOR

$$\sigma = \begin{bmatrix} \sigma_{xx} & \tau_{xy} & \tau_{xz} \\ \tau_{xy} & \sigma_{yy} & \tau_{yz} \\ \tau_{xz} & \tau_{yz} & \sigma_{zz} \end{bmatrix}$$

Symmetry follows moment equilibrium.
Principal stresses are the eigenvalues.

TRACTION ON A PLANE

$$\mathbf{t}(\mathbf{n}) = \sigma \mathbf{n}$$

$$\sigma \mathbf{n} = \mathbf{n} \cdot \mathbf{t}$$

$$\boldsymbol{\tau} = \mathbf{t} - \sigma \mathbf{n}$$

The plane normal $\mathbf{n}$ controls the
resolved normal and shear
components.

ROTATE INTO BOREHOLE AXES

$$\sigma^b = \mathbf{A} \sigma^G \mathbf{A}^T$$

Rows of $\mathbf{A}$ are the borehole basis
vectors expressed in global
coordinates.

$$\text{Check: } \mathbf{A} \mathbf{A}^T = \mathbf{I}$$

Transform the tensor first. Then apply the borehole-wall equations in the local coordinate system.

Biot effective stress

Pore pressure changes the stress carried by the porous skeleton

Compression-positive form

$$\sigma' = \sigma - \alpha P_p$$

For isotropic poroelasticity

$$\alpha = 1 - K_d / K_s$$

Term	Meaning	Evidence
σ	total stress tensor	density integration, stress tests
P_p	connected-pore pressure	pressure tests, calibrated prediction
α	Biot coefficient	drained bulk modulus and grain modulus
σ'	skeleton stress used by the selected model	calculated consistently

α is measured or constrained. Setting $\alpha = 1$ without evidence can bias stress-path and failure calculations.

Kirsch equations at the borehole wall

Elastic wall stresses vary with azimuth even when the far-field stresses are uniform

WALL CONDITIONS

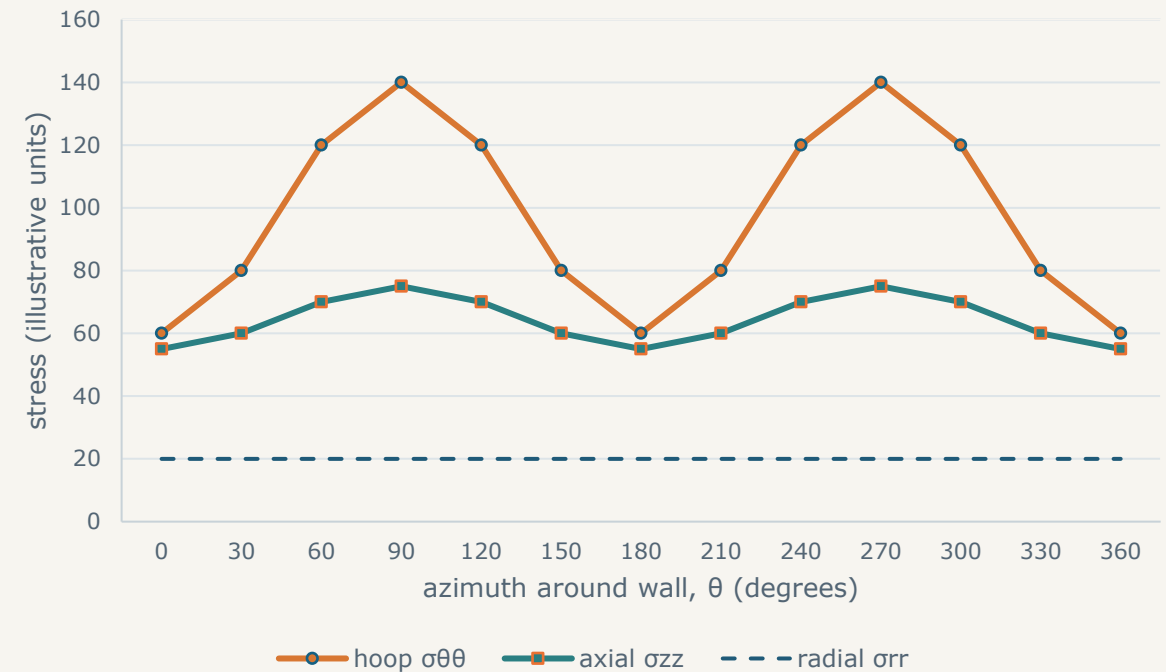
$$\sigma_{rr}(a, \theta) = P_w$$

$$\tau_{r\theta}(a, \theta) = 0$$

$$\begin{aligned} \sigma_{\theta\theta}(a, \theta) = & \sigma_x + \sigma_y \\ & - 2(\sigma_x - \sigma_y) \cos 2\theta \\ & - 4\tau_{xy} \sin 2\theta - P_w \end{aligned}$$

$$\begin{aligned} \sigma_{zz}(a, \theta) = & \sigma_z \\ & - 2\nu(\sigma_x - \sigma_y) \cos 2\theta \\ & - 4\nu\tau_{xy} \sin 2\theta \end{aligned}$$

Illustrative wall stresses



Assumptions shown: circular hole, homogeneous isotropic elasticity, plane strain, no thermal or poroelastic correction.

Inclined-borehole stress transformation

Trajectory changes the normal and shear stresses resolved around the hole

LOCAL BASIS

$\mathbf{z}^b$ along the borehole axis

$\mathbf{x}^b$ reference direction in the cross-section

$\mathbf{y}^b$ completes a right-handed basis

$$\boldsymbol{\sigma}^b = \mathbf{A} \boldsymbol{\sigma}^G \mathbf{A}^T$$

CALCULATION SEQUENCE

- 01 Define azimuth, inclination and axis convention
- 02 Rotate the full far-field stress tensor
- 03 Apply the wall solution with local shear terms
- 04 Calculate principal wall stresses and failure index
- 05 Map predicted failure back to image-log coordinates

WHEN AZIMUTH RULES FAIL

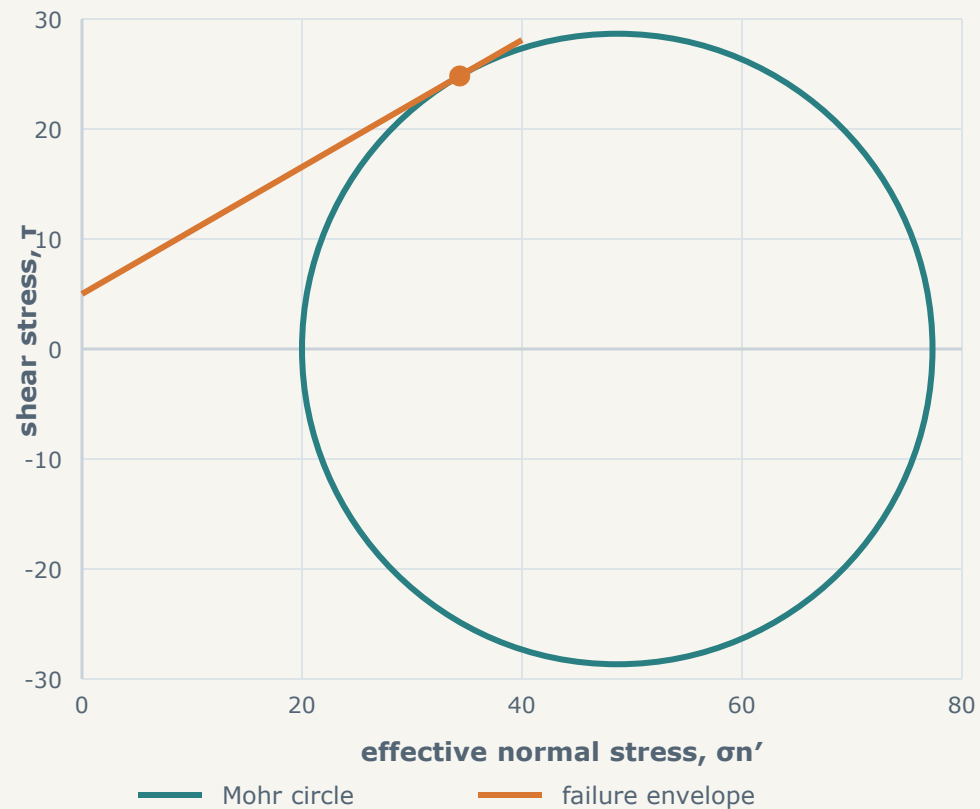
- Local shear stresses may rotate the principal wall stresses.
- Breakout and DITF traces may become oblique or en echelon.
- Cooling, anisotropy and weak planes can change the trace geometry.
- A 3D forward model should reproduce both location and morphology.

The image-log trace belongs to the borehole surface, not directly to the geographic stress axes.

Mohr circle

Principal effective stresses define the normal and shear stress available on every plane

Mohr circle at incipient Mohr-Coulomb failure



$$C = (\sigma_1' + \sigma_3') / 2$$

$$R = (\sigma_1' - \sigma_3') / 2$$

Example

$$\begin{aligned}\sigma_3' &= 20 \text{ MPa} \\ \sigma_1' &= 77.3 \text{ MPa} \\ c &= 5 \text{ MPa} \\ \phi &= 30^\circ\end{aligned}$$

Tangency marks the predicted failure state for this criterion and parameter set.

Mohr-Coulomb criterion

Cohesion and friction angle define a linear shear-failure envelope

Plane form

$$\tau = c + \sigma n' \tan \varphi$$

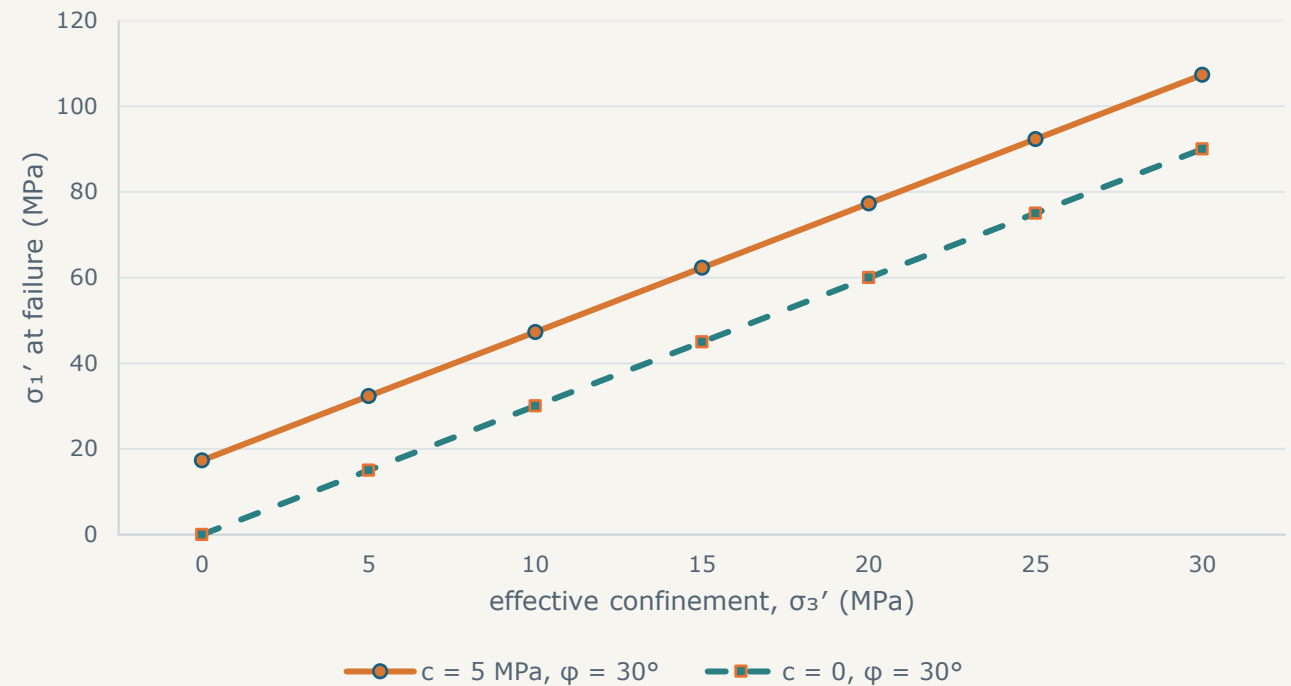
Principal-stress form

$$\sigma_1' = C_0 + q \sigma_3'$$

$$q = (1 + \sin \varphi) / (1 - \sin \varphi)$$

$$C_0 = 2c \cos \varphi / (1 - \sin \varphi)$$

Confinement raises predicted compressive strength



Fit c and φ over the stress range that controls the wellbore. A linear envelope can misrepresent curvature or σ_2 effects.

Failure-criterion comparison

Criterion choice should follow the available tests, stress path and decision sensitivity

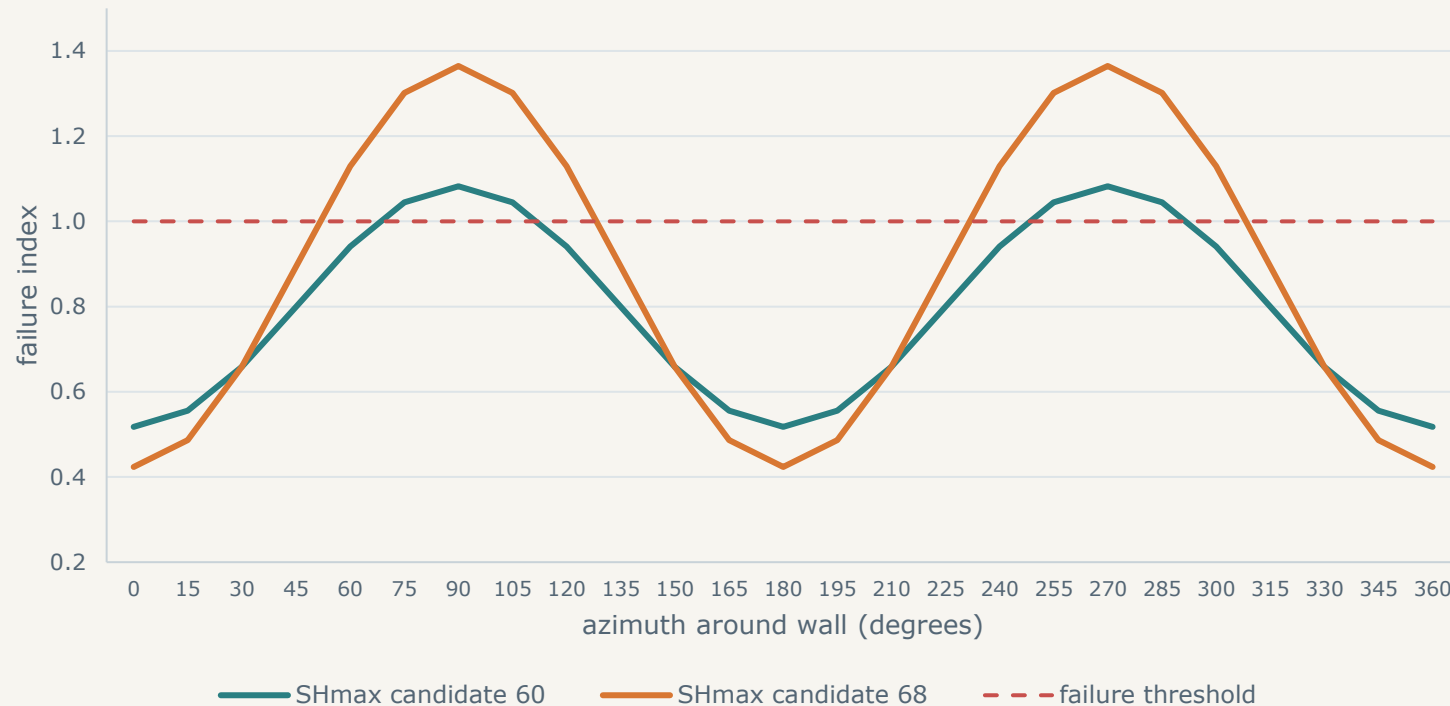
Criterion	Envelope	Uses σ_2 ?	Calibration strength	Main caution
Mohr-Coulomb	linear in σ_1 - σ_3	No	transparent c and ϕ	may miss curvature and σ_2 effect
Hoek-Brown	nonlinear in σ_1 - σ_3	No in standard form	fits confinement curvature	intact and rock-mass forms differ
Mogi-Coulomb	linear in octahedral shear vs mean stress	Yes	extends Coulomb to polyaxial states	parameter mapping and data range matter
Modified Lade	3D invariant surface	Yes	often useful for polyaxial rock strength	less intuitive parameters
Drucker-Prager	smooth pressure-dependent cone	Yes through invariants	convenient numerically	inscribed and circumscribed forms differ

Compare model misfit and predicted mud-window sensitivity before choosing added complexity.

Breakout width as a stress-magnitude constraint

Width becomes quantitative only after the wall-stress and rock-strength assumptions are fixed

Illustrative failure index around the borehole



INVERSION INPUTS

- Directional breakout width from an image log
- Well trajectory and borehole orientation
- Pore pressure and actual wellbore pressure
- Rock strength, anisotropy and failure criterion
- Thermal state and exposure history

$$wBO = 2 \theta B$$

Fit the SHmax family that reproduces the measured failed-zone width and morphology.

DITF interpretation in inclined wells

The trace geometry depends on the transformed three-dimensional wall-stress field

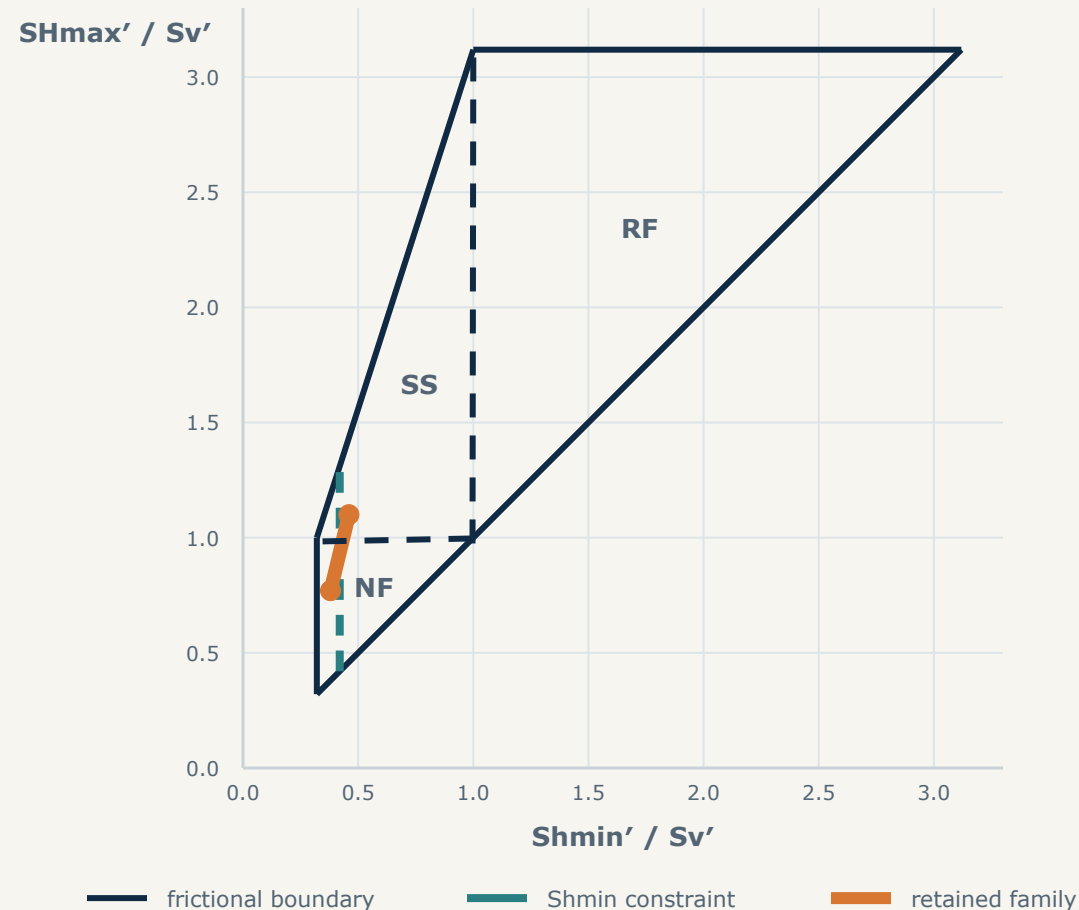
Factor	Simple vertical-well case	Inclined-well interpretation
Stress transformation	principal axes may align with the hole	local normal and shear components generally coexist
Trace orientation	axial traces may align with SHmax	traces can be oblique, curved or en echelon
Pressure and cooling	reduce minimum wall stress	effect varies with trajectory and circulation history
Fabric and fractures	secondary in the ideal model	may redirect initiation and propagation
Stress inversion	azimuth rule can be useful	fit a 3D forward model to the complete trace pattern

Tensile initiation screen: minimum principal wall stress + thermal term $\leq -T_0$

Stress polygon

Frictional faulting bounds the admissible horizontal-stress family but rarely selects one solution

Normalized frictional admissibility, $\mu = 0.6$



BOUNDING RELATION

$$\sigma_1' \leq q \sigma_3'$$

$$q = [\sqrt{(\mu^2 + 1)} + \mu]^2$$

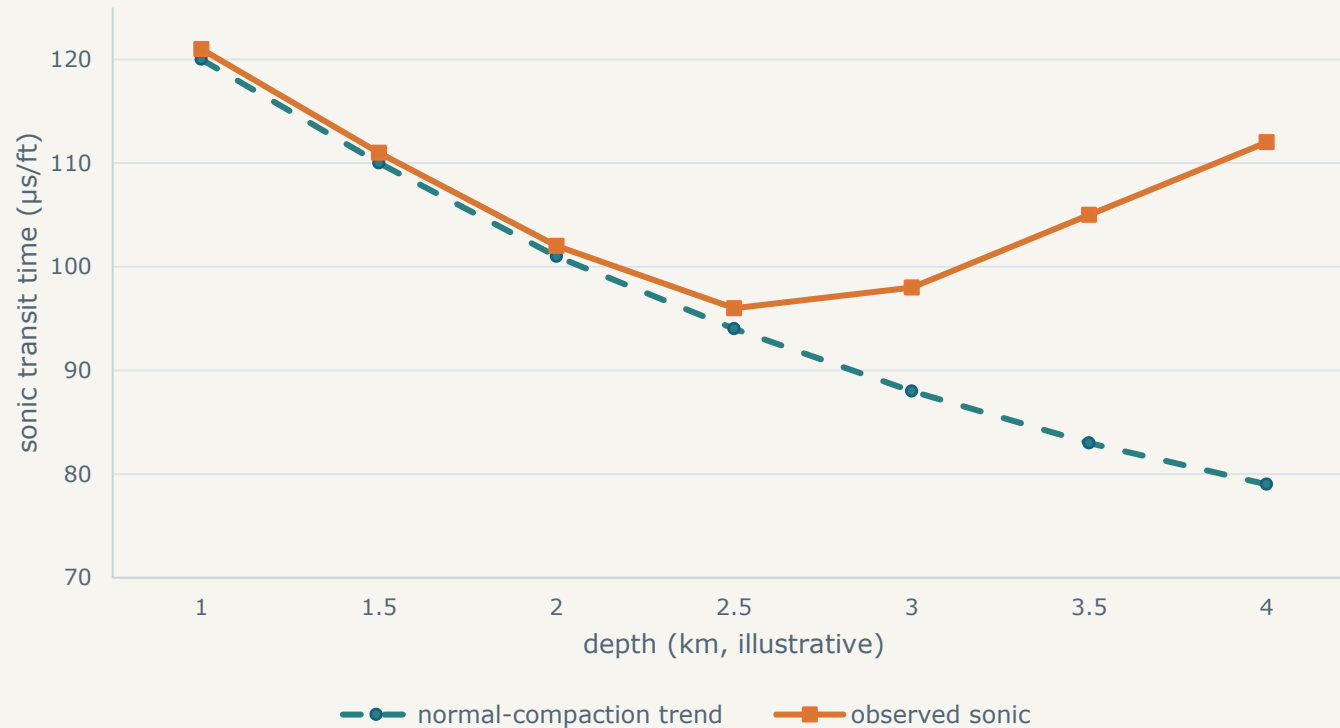
- S_v and P_p set the normalization.
- A stress test can narrow Sh_{min} .
- Breakout or DITF geometry adds another band.
- The intersection remains a family until evidence narrows it.

The colored constraints are illustrative, not field measurements.

Eaton pore-pressure concepts

Departure from a calibrated normal-compaction trend can indicate undercompaction overpressure

Illustrative sonic departure from the normal trend



Generic sonic form

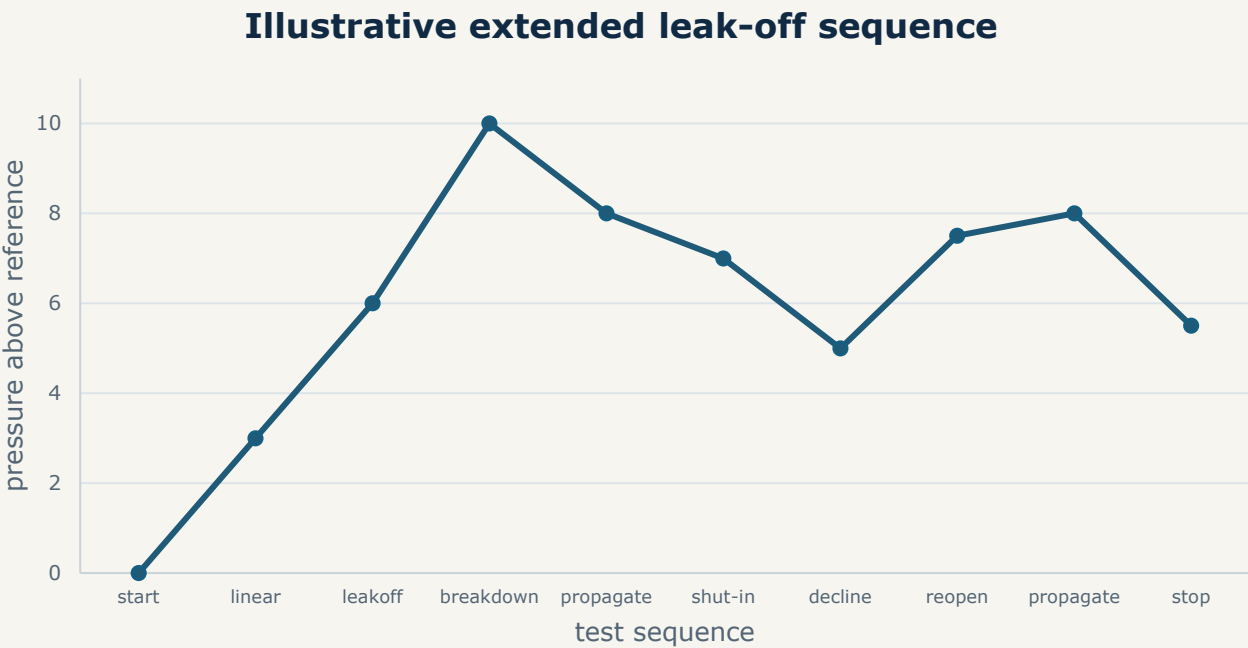
$$P_p = S_v - (S_v - P_n) (\Delta t_n / \Delta t_{obs})^n$$

CALIBRATION CHECKS

- Lithology and log quality
- Trend interval and exponent n
- Temperature, cementation and diagenesis
- Unloading or other overpressure mechanisms
- Direct pressure and drilling calibration

FIT, LOT and XLOT interpretation

Test names summarize procedures, while the pressure-rate-volume record carries the evidence



Test	Typical evidence	Does not automatically prove
FIT	integrity to achieved pressure	leakoff, breakdown or closure
LOT	departure from linear pressurization	minimum stress without interpretation
XLOT	breakdown, propagation, shut-in and reopening cycles	one unique stress value under every analysis method

Preserve depth datum, fluid density, temperature, pump rate, cumulative volume, pressure location and the interpretation rule.

ECD, surge and swab

Actual bottomhole pressure moves as circulation and pipe motion change annular pressure

Circulating pressure

$$\text{BHP}_{\text{circ}} = \rho_{\text{MW}} g \text{ TVD} + \Delta P_{\text{ann}}$$

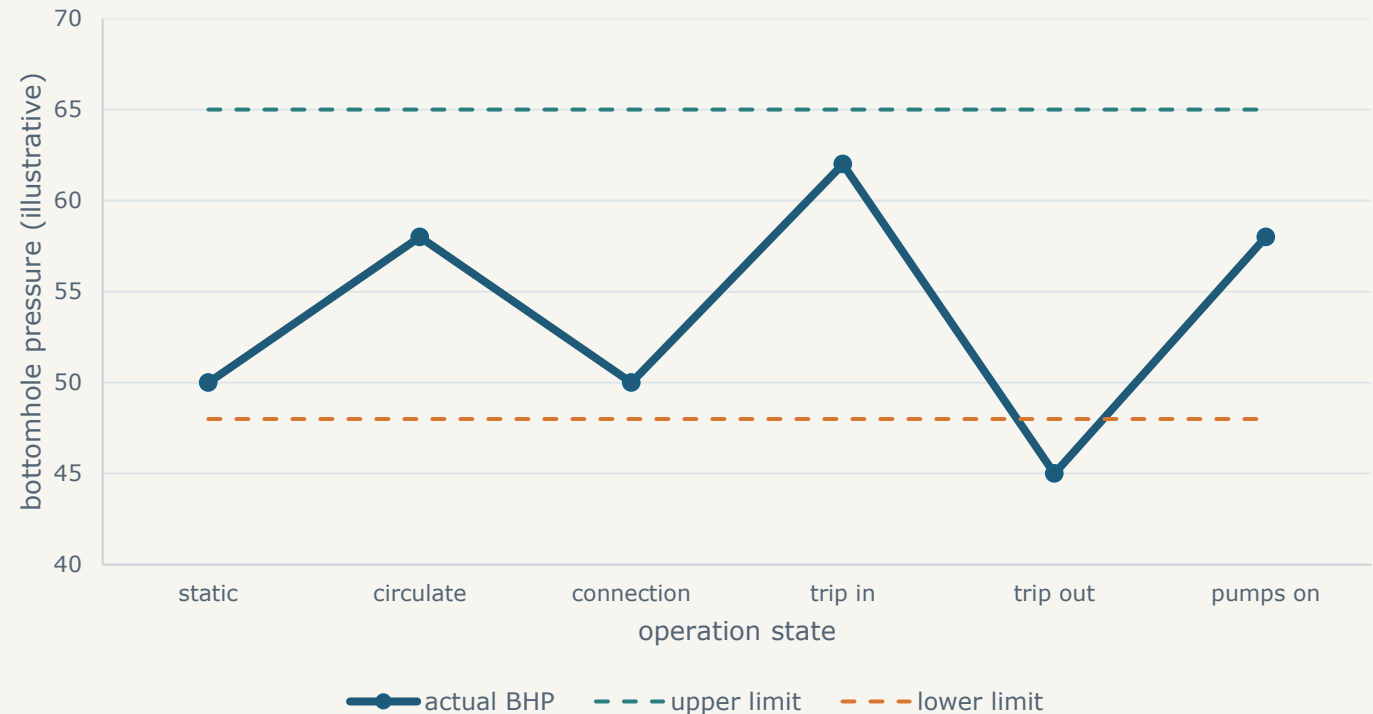
Density-equivalent form

$$\text{ECD} = \rho_{\text{MW}} + \Delta P_{\text{ann}} / (g \text{ TVD})$$

Surge adds pressure during downward pipe motion.

Swab subtracts pressure during upward motion.

Illustrative dynamic bottomhole-pressure path



A static mud weight can sit inside the window while swab or surge crosses a limit.

Shale anisotropy, time and fluid interaction

The lower stability limit can evolve after exposure even when the applied pressure remains unchanged

Mechanism	Possible rig signature	Laboratory evidence	Model response
Bedding and fabric	directional cavings or trajectory sensitivity	oriented strength, stiffness and wave velocity	anisotropic properties or weak-plane criterion
Pore-pressure equilibration	delayed change after new exposure	permeability, poroelastic response, time scale	coupled pressure diffusion
Creep and delayed damage	progressive enlargement during static time	creep compliance and stress-path tests	time-dependent strength or deformation
Fluid chemistry	response changes with mud system or salinity	activity-controlled exposure and retained-strength tests	chemical potential or calibrated weakening

Match specimen orientation, saturation, chemistry, temperature, stress path and exposure time to the field question.